

An open-source Thermo-hydraulic Uniphase Advection and Convection Solver for Salt Flows (TUAS)

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ABSTRACT

Due to a lack of open-source GUI simulators and corresponding system level simulation libraries for Gen IV Reactors such as the Fluoride Salt Cooled High Temperature Reactor (FHR), a systems level library for single phase flow named “TUAS” was written in the Rust which was then used to construct the first educational simulator of the Compact Integral Effects Test (CIET). In this work, we successfully performed a verification and validation (V&V) study on TUAS using idealized natural circulation simulation results from the System Analysis Module (SAM) simulation of CIET, experimental data of a HITEC salt to YD-325 oil Shell and Tube Heat Exchanger (STHE) as well as experimental forced circulation data from CIET. TUAS was shown to match SAM results well for the natural circulation verification study and experimental data well for forced circulation validation studies. Real-time simulation capabilities on consumer laptops for forced circulation were also demonstrated for the Open Source TUAS library using the CIET Educational Simulator with potential for validated real-time natural circulation in future work. Moreover, TUAS could be coupled with multiphase solvers and neutronics feedback in future to build real-time 2D and 3D reactor simulators for both research and outreach. These efforts contribute to a richer open-source nuclear reactor safety simulation ecosystem built in modern system programming languages such as Rust. Such an ecosystem would provide tools for a more conducive environment in education and research for nuclear engineering.

1. Introduction

Interest in nuclear power [1] to curb carbon emissions (also known as “net zero” [2]) has been growing over the years. To achieve this goal while addressing past safety concerns [3], Generation IV (Gen IV) nuclear power plants, such as the Fluoride Salt Cooled High Temperature Reactor (FHR), were designed with passive safety in mind [4]. However, the development process of Gen IV reactors like the FHR is often cost prohibitive [5]. Moreover, lack of public trust [6] in and awareness of the safety of nuclear power has often introduced additional barriers to widespread deployment of Gen IV reactors [7,8]. These issues are especially important to address in context of countries wishing to explore usage of nuclear technology for power and industrial purposes such as those found in ASEAN including the Philippines, Malaysia, Indonesia, Thailand and Singapore. While ASEAN countries have expressed interest in usage of nuclear power, projects have never taken off or were abandoned halfway due to public safety concerns or

economic reasons [9]. To ameliorate these twin issues, reactor simulators can be used. To speed up FHR development, system level simulation codes validated using integral effects tests (IETs) have been used. Kairos Power has used these tools to great effect within its rapid iteration approach [10] to develop the Kairos Power FHR (KP-FHR). In a similar manner, reactor simulators with a graphical user interface (GUI) have been used for operator training [11] and public outreach and education [12,13].

Real-time reactor GUI simulators, while useful for research, education and outreach, are often proprietary, which limits their reach in achieving the desired outcomes in these areas. For example, real time reactor simulators were used for public outreach in Virginia Commonwealth university. The transients were calculated using best estimate system codes such as RETRAN-3D while the GUI was provided using student developed Labview code [14]. RETRAN and Labview are not open-source and therefore not available to the public. Like the 2D simulators, 3D reactor simulators have also been developed for use in

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visualization using the Unity3d engine [15]. These 3D simulators, especially virtual reality and augmented reality simulators, have been used in attempts to reduce costs when it comes to staff training, education, and plant construction [16]. Unfortunately, the source code for these simulators are not readily available. In a similar manner, GUI simulators such as PCTTRAN used in research to study steam generator tube rupture [17] are closed source and proprietary. Likewise, tools used to simulate and develop the FHR, such as SAM [18,19] and RELAP [20, 21] are often proprietary. For ASEAN countries considering nuclear power, access to these codes and GUI simulators is not guaranteed, which inevitably slows down progress in achieving net zero goals.

In fact, there are limited choices for open-source systems level simulation of FHRs in 2024. Moreover, Open Source simulators with GUIs are even fewer. The most notable system level open-source tools for Molten Salt Reactor (MSR) and FHR simulation are within the Modelica and Transform ecosystem [22–24]. However, Transform relied upon Dymola, which is not open-source [25]. Therefore, Pathirana et al. developed the scalable modular dynamic molten salt reactor (SMD-MSR) system which could operate in real-time independently from Dymola [25,26] using OpenModelica. Nevertheless, open-source options for Graphical User Interface (GUI) tools for MSR simulators within the Modelica ecosystem are severely limited with OMEdit [27] being one of few options available. For constructing real-time 3D simulators, the Modelica ecosystem is not as suitable as it was never designed with this purpose in mind. Therefore, where 2D and 3D open-source GUI simulators for FHRs are concerned, there is a lack of tools available.

To address this lack, an educational simulator prototype with a 2D GUI of the Compact Integral Effects Test (CIET), a scaled IET of the FHR, was constructed. This is a first step towards constructing an open-source ecosystem of 2D and 3D FHR simulators and hopefully Gen IV reactor simulators for education, outreach and research. The simulator had to be written from the ground up using code that was high performance and relatively quick to develop in. The Rust programming language was chosen in preference to Fortran, C and C++ due to its ease with memory management and dependency management [28,29]. Moreover, the Rust programming language was more well suited to building GUI simulators as it has several choices of 2D and 3D GUI libraries such as the 2D GUI engine, egui [30], and the 3D game GUI engines Bevy [31] and Fyrox [32]. The latter libraries are potentially suitable choices for building 3D simulators in future work. For this work, the 2D GUI was written in Rust using the existing GUI library, egui [30]. However, a systems level thermal hydraulics library required to simulate CIET in real-time was still lacking. To address this lack, a single-phase systems level thermal hydraulics library based on the Boussinesq approximation was written in Rust and is now presented in this work. This solver is called the Thermo-hydraulic Uniphase solver for Advection and Convection in Salt flows (TUAS), named after the Tuas industrial region in the west of Singapore. The construction of TUAS then enabled the development of the CIET Educational Simulator, which works natively on Windows, Linux and MacOS.

Early development for the TUAS library, was motivated by the need designed for real-time simulation of the Compact Integral Effects Test (CIET) and molten salt systems for Simulated Neutronics Feedback Controller Development [33,34]. However, early iterations, then known as the Boussinesq Solver within the thermal_hydraulics_rs library, were incomplete as it lacked demonstrated capability to solve natural circulation flows and forced circulation loops. Nevertheless, it demonstrated real-time simulation capability also allowed for its use in remote operation studies which is of interest for advanced reactor control [35] and digital twin applications. In such a setup, the TUAS library would solve thermal hydraulics equations for CIET in real-time, while emulating the communication of CIET with its Supervisory Control and Data Acquisition (SCADA) software via the Open Platform Communications Unified Architecture (OPC-UA) server and client interface. This capability was demonstrated in previous work using open-source OPC-UA and 2D GUI

libraries written in the Rust programming language [34]. However, the TUAS library was incomplete then, as it was only proven to be able to simulate isothermal Therminol VP-1 flow within CIET [33] and CIET's heater [34]. In this work, the TUAS library was completed and has demonstrated capability to simulate forced and natural circulation flows. Preliminary verification and validation (V&V) of TUAS was done in previous work as a PhD Dissertation [34]. This included isothermal pressure drop tests (page 162 of dissertation), lumped capacitance transient simulations (page 127–135 of dissertation), semi infinite medium transient conduction (page 136–140 of dissertation), conjugate heat transfer (CHT) of a lumped capacitance model of CIET's heater (page 141–154) and CHT simulation of a nodalised model of CIET's heater without insulation (page 155–197) [34] with a mesh refinement study. This previous V&V work was incomplete, therefore we performed V&V studies in the context of natural circulation within CIET, as well as shell and tube heat exchangers (STHE). This is part of a larger effort to enable TUAS to simulate steady state natural circulation in coupled loops like SAM [36]. In this work, the TUAS simulation results for the idealized DRACS loop in CIET matched well with SAM. Additionally, the TUAS simulation results for a HITEC salt to oil heat exchanger matched well with experimental data. Moreover, this work demonstrates the ability of the TUAS library to simulate the isolated DRACS loop and STHE in real-time using laptops computers. The TUAS library is published on Github along with its documentation [37].

The success of the V&V studies then paved the way for construction of the first open-source real-time 2D GUI educational simulator of CIET written using the TUAS and egui libraries presented in this work. The source code of this simulator is published as an example of TUAS within its Github repository [37]. Key features of this simulator are presented, including (1) the ability of the user to control the heater power with step inputs for time domain step response tests and sinusoidal inputs for frequency response tests, (2) the ability to visualize temperature profiles around CIET using pipe components with colour changing according to fluid temperature, (3) the ability to adjust outlet temperatures of the fan based coolers using filtered Proportional, Integral and Derivative (PID) controllers, (4) the ability to conveniently perform transient validation tests, visualize and plot data using real-time graphs and csv values, and (5) the ability to compile natively on Windows, Linux and MacOS. The step response validation tests performed in item (4) also serves to validate the existing insulated heater model programmed in the CIET educational simulator using TUAS.

This paper first covers coupled natural circulation flows in CIET as the key motivation for the DRACS loop code to code verification and shell and tube heat exchanger (STHE) validation for this work in Part 2.1. Part 2.2 presents conservation equations used for the STHE and DRACS Loop. Part 3.1 then discusses discretization of equations and part 3.2 discusses the solution procedure. Part 3.3 discusses code implementation details. Part 3.4 describes component V&V. Simulation details of the CIET educational simulator are described in Part 3.5. The results for the isolated DRACS loop verification tests are presented in Part 4.1. Thereafter, the STHE validation tests are presented in Part 4.2. Key features of the CIET Educational Simulator, along with its step response validation test are presented in Part 4.3. Part 5 is where the conclusions are presented.

2. Theory

2.1. Natural circulation in CIET

CIET is an integral effects test (IET) using the simulant fluid Dowtherm A to simulate Li_2BeF_4 (FLiBe) within the FHR [21,38]. Fig. 1 presents the model to simulate natural circulation in CIET:

In Fig. 1, the electrical heater (component 1), simulating decay heat in FHRs, adds heat to the primary loop in the heater branch (components 4 to 18). Hot Therminol VP-1, also known as Dowtherm A, rises in the heater branch and is cooled in the Direct Reactor Auxiliary Cooling

of coupled natural circulation in literature was performed in two steps. Firstly, an isolated DRACS loop without parasitic heat loss was simulated and compared with an analytical solution [36] to verify that components were simulated correctly. Secondly, the coupled natural circulation validation was performed by comparing simulation to experimental data of mass flowrates in both primary and DRACS loop [36]. We intend to follow the two step approach similar to how V&V for SAM was performed using CIET.

In the first step, code to code verification is done using the isolated DRACS loop without parasitic heat loss, as a reference. The spatial discretization was determined using the SAM nodalisation scheme found in page 23 of Zou's SAM publication [18]. Additionally, the shell and tube heat exchanger (STHE) code is validated using experimental data. For the DHX, experimental calibration data was not available in literature. Therefore, code simulating a STHE is validated using a salt to oil STHE [40,41], shown in Fig. 2, so we can have confidence that the DHX is simulated correctly in future work for coupled natural circulation loops.

2.2. Physical equations in natural and forced circulation

To simulate both natural and forced circulation, the mass, momentum and energy balances were programmed into the TUAS library. The TUAS library is also based on the Boussinesq approximation [42] which simplified the thermal hydraulics equations. This set of equations is much simpler than the multiphase equations RELAP and SAM solve, and is therefore best suited to real-time single phase simulation of FHRs. While it does not allow for salt freezing simulations like SAM [43], the simplification allowed for faster solver development and solver execution.

For mass balances, $\dot{m}$ was assumed to be the same in all components within the same branch. For n parallel branches, the mass balance is shown in (1) [33]:

$$\dot{m}_{\text{total}} = \sum_i^n \dot{m}_{\text{branch},i} \quad (1)$$

For momentum balance, we compute ΔP across each branch, where

each branch contains n components, using (2):

$$\Delta P_{\text{change branch}} = - \sum_i^n \frac{1}{2} \frac{\dot{m}_{\text{branch},i}^2}{\rho_i A_{xs,i}^2} \left(f_{\text{darcy}} \frac{l}{d_h} + K \right)_i + \sum_i^n \rho_i g z_i + \sum_i^n \Delta P_{\text{source},i} \quad (2)$$

Where ρ_i is the average density computed using a mass averaged temperature T_i . Supposing that a fluid pipe is discretised into 1 control volumes with mass m_j and temperature T_j , the ρ_i is calculated using Equation (3):

$$T_i = \frac{1}{m_i} \sum_j^l T_j m_j \quad (3)$$

Where $m_i = \sum_j^l m_j$. In natural circulation mode, $\Delta P_{\text{source},i} = 0$ Pa as no pumps are active to provide pressure head to circulate fluid. The only source of pressure increase comes from the difference in $\sum_i^n \rho_i g z_i$ between each branch. This must be equal to $\sum_i^n \frac{1}{2} \frac{\dot{m}_{\text{branch},i}^2}{\rho_i A_{xs,i}^2} \left(f_{\text{darcy}} \frac{l}{d_h} + K \right)_i$ in natural circulation mode. For a full natural circulation loop, two parallel branches are used to simulate the hot and cold legs of the loop respectively. The 1D energy balance is given in Equation (4) [34]:

$$\rho_i A_{xs,i} \frac{\partial h_{\text{enthalpy}}}{\partial t} = - \rho_i A_{xs,i} u \frac{\partial h_{\text{enthalpy}}}{\partial x} + \sum_j q_j P_{\text{heated},j} \quad (4)$$

Equation (4) states that the change in enthalpy in a differential volume is the sum of advection interactions and j heat flux interactions acting on that differential volume. For a tube-shaped wall, $P_{\text{heated},j}$, the heated perimeter, is simply πd_i .

3. Calculation, materials and methods

3.1. Energy balance discretization

To solve equation (4), it must first be discretised. In contrast to SAM's

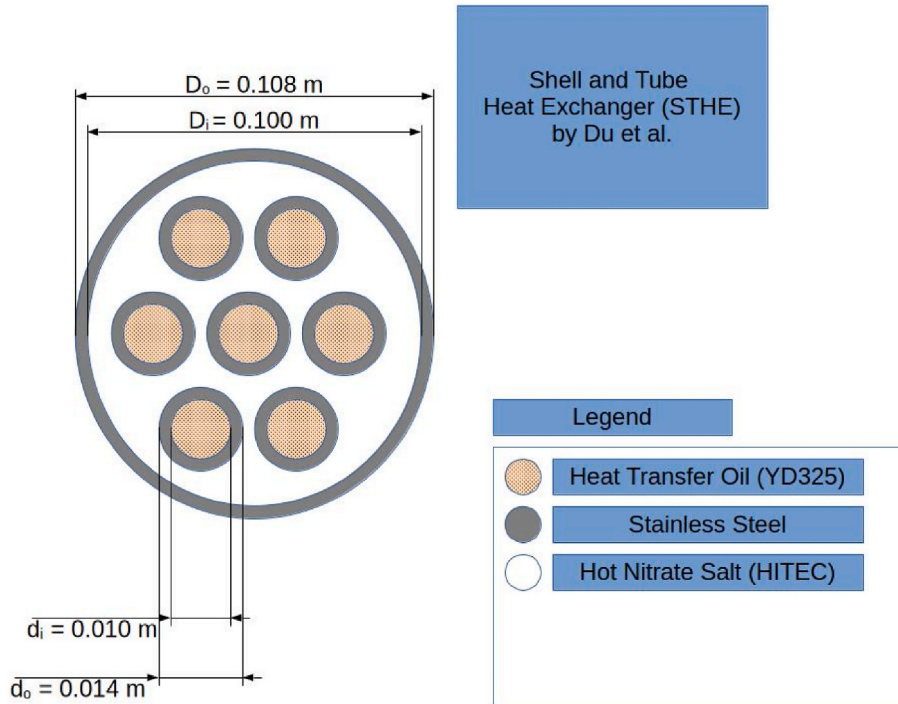


Fig. 2. Simplified Cross Section of Du et al.'s STHE (not drawn to scale, and number of tubes is not representative of Du's STHE Setup to simplify drawing) Additionally, validation with a salt and oil STHE improves confidence that the TUAS Solver can simulate molten salt flows for FHRs.

Finite Element Method (FEM) based on MOOSE [43], the TUAS library relies on a one-dimensional finite volume approach. For illustration purposes, it is useful to consider how this is done for a shell and tube heat exchanger (STHE) such as Du et al.'s heat exchanger [40]. A simplified schematic of such a heat exchanger is provided in Fig. 3.

Fig. 3 shows the typical layout of a STHE conceptualized as control volumes connected by thermal resistances. For the tube side fluid, using a semi-implicit time coupling scheme, each well-mixed control volume has the discretised form of Equation (4) resulting in Equation (5):

$$\rho_i c_p^t (T_x^t) A_{xs,x} \Delta x \frac{T_x^{t+\Delta t} - T_x^t}{\Delta t} = -\rho_i c_p^t (T_x^t) A_{xs,x} u (T_x^{t+\Delta t} - T_{x-\Delta x}^{t+\Delta t}) + \sum_j q_j^t A_j \quad (5)$$

Equation (5) uses the relatively simple first-order upwind method for its energy equation, which is diffusive but avoids oscillatory behaviour found in second order schemes [44]. For first order schemes, numerical diffusion errors. However, they can be tolerated if the errors are small enough. When quantifying what errors are significant in V&V efforts, it is important to refer to errors typical of code-to-code comparisons as well as experimental validation studies. For errors typical in V&V, we can refer to studies by Zweibaum et al. They reported that RELAP5 agreed with CIET experimental data (natural circulation mass flowrates) to within 1 % whilst the FANCY natural circulation code agreed with the same experimental data to within 3 % [20]. Based on this information, we can perform a crude estimate that the discrepancy between RELAP and FANCY is around 2–4 %. Therefore, for code-to-code verification, a discrepancy of around 2–4 % may be considered reasonable. Whereas for experimental data, a 3 % deviation of mass flowrates the isolated DRACS is reasonable. For the isolated DRACS loop, the discretization scheme used in the SAM model, found on page 23 of Zou's publication [18], was used. The natural circulation mass flowrates were compared against the analytical solution used to benchmark SAM [36].

For STHEs, the experimental Nu deviated from that of the empirical

correlation provided by about 8 % [40]. Therefore, when performing validation studies in modelling Du's STHE, an error of no more than 8 % is shown that the first order scheme is acceptable.

Besides ensuring that deviations fall within this acceptable range, it is also important to perform benchmarks where the code is compared against high fidelity accurate solutions [45] so that there is confidence that TUAS works outside the DRACS loop and STHE simulations. As the first order diffusive scheme would affect transient conduction the most, it is important to benchmark transient conduction of TUAS using an analytical solution. This was performed to a limited degree in a semi-infinite medium transient conduction benchmark reported on page 136–140 in a previous PhD dissertation [34] for earlier versions of TUAS where the same first order scheme was used, thus verifying that the conduction solvers were correctly coded. This gives additional assurance that the 1D conduction solvers were coded correctly.

Now, equation (5) assumes that fluid flows in the positive x direction so that the enthalpy flowing out of the control volume depends on $T_x^{t+\Delta t}$. If the fluid is flowing in the negative x direction, enthalpy flowing into the control volume depends on $T_{x+\Delta x}^{t+\Delta t}$ rather than $T_{x-\Delta x}^{t+\Delta t}$. Additionally, semi-implicit time discretization in equation (5) was used for the control volumes because it allowed for faster computation time compared to a fully implicit coupling scheme, where larger matrices had to be constructed and solved in $\mathbf{A} \vec{x} = \vec{b}$. This solution scheme also allowed for a higher degree of parallel computing to allow for solution procedures in real-time [34]. For the tube side fluid, the $q_j^t A_j$ terms through the inner tube wall were calculated via a thermal conductance or resistance method using temperatures from the last timestep. Other sources of heat flux include axial thermal conduction within the tube, which for high Pe flows, is negligible relative to axial convection [46].

For the solid tubes, the advection terms are set to zero. An example for the leftmost control volume of solid inner tube in Fig. 3 is shown in Equation (6):

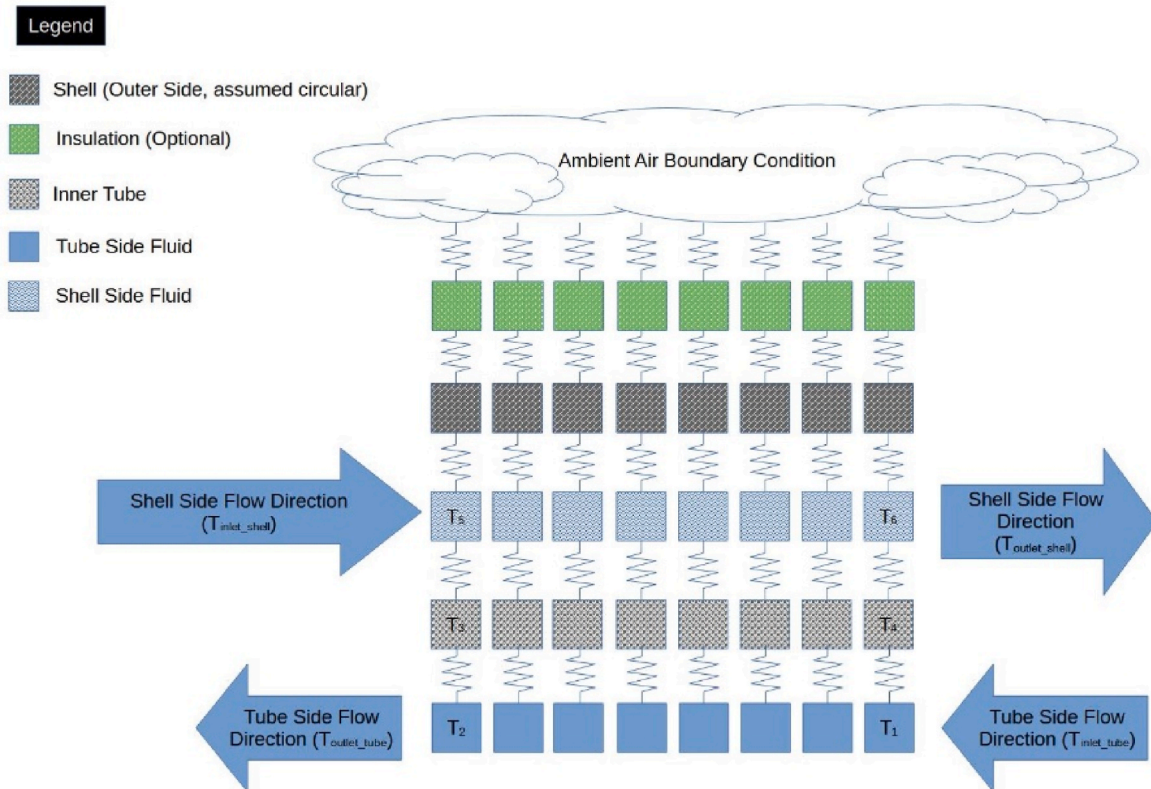


Fig. 3. Control Volume Layout in Du et al.'s Heat Exchanger.

$$\rho_i c_p^t (T_3^t) A_{xs,3} \Delta x \frac{T_3^{t+\Delta t} - T_3^t}{\Delta t} = - \frac{\lambda_{inner tube}^t A_{XS}}{\Delta x} (T_3^{t+\Delta t} - T_{3+\Delta x}^{t+\Delta t}) + \dot{Q}_{boundary} + \frac{-(T_3^t - T_5^t)}{\frac{\ln(d_o/d_{mid})}{2\pi\lambda_{inner tube}^t (T_3^t)\Delta x} + \frac{1}{h_{shell}\pi d_o\Delta x}} + \frac{-(T_3^t - T_2^t)}{\frac{\ln(d_{mid}/d_i)}{2\pi\lambda_{inner tube}^t (T_3^t)\Delta x} + \frac{1}{h_{tube}\pi d_o\Delta x}} \quad (6)$$

Where with reference to Fig. 2, $d_{mid} = \frac{d_o + d_i}{2}$, $h_{tube,j}$ is computed using the Gnielinski correlation [40,47]. For control volumes not at the boundary, $\dot{Q}_{boundary}$ is 0 W and an additional axial conduction term is added. Even when simulating Du et al.'s STHE, $\dot{Q}_{boundary} = 0$ W as an adiabatic boundary condition was used. Now, Equation (6) also presumes that only one radial node is used for heat transfer across tube walls. If more nodes are desired, TUAS is flexible enough to allow the user to add additional radial nodes to construct STHE simulation objects. For this work, only one radial node is used to represent tube walls. To ascertain if this is sufficiently accurate, Du's STHE simulation must predict Nu to within the 8 % tolerance mentioned earlier.

On the shell side of a single pass STHE, h_{shell} is calculated using an empirical Nusselt number correlation. Du et al.'s empirical correlation is shown in Equation (7) [40]:

$$Nu_{shell} = 0.04318 (Re_f^{0.7797} - 280) Pr_f^{0.4} \left(1 + \left(\frac{D_e}{l} \right)^{2/3} \right) \left(\frac{Pr_f}{Pr_{wall}} \right)^{0.25} \quad (7)$$

For the shell side within Du et al.'s STHE, it must be noted that there were 19 parallel tubes [40]. To account for this, the STHE allows for N_t parallel tubes to be represented by one single tube during calculations. Additionally, in Du et al.'s STHE simulation, no parasitic heat loss is simulated. This approach was followed in this work. For generic STHEs, control volumes representing a metallic outer tube and insulation layer would be programmed as shown in Fig. 3. If one desires to simulate parasitic heat loss, a third user programmable Nu correlation would be required to program the thermal resistance to the outer shell in Fig. 3 and specify if the STHE has any form of insulation. The outer shell and insulation layers would be programmed in a manner like the inner shell shown with discretised energy balance shown in Equation (6).

3.2. Solution procedure

3.2.1. Operator splitting

To facilitate real-time calculations, the TUAS library solves the mass and momentum balances separately from the energy balances at each timestep. This procedure, known as operator splitting [48] can be visualized in Fig. 4 [33]:

Operator Splitting allows for a different computing thread to solve the mass and momentum balance and the energy balance. The mass, momentum and energy balances each use fluid properties and mass flowrates at the previous timestep to calculate the new temperatures, fluid properties and mass flowrates at the next timestep. Thus, the solution is semi-implicit as compared to a fully implicit solver in SAM [43]. This was done to make it easier to introduce parallelism if needed to enable real-time calculations.

To illustrate this, let us consider forced circulation along the shell side of the heat exchanger. Suppose now there is a fixed pressure drop across the shell side, and that fluid properties and flowrates at the last timestep are known. Assuming a horizontal STHE, a constant pressure change, and that there is no internal pressure source, Equation (2) becomes simplified to equation (8), which is used to compute the mass flowrate within the shell at the next timestep:

$$\Delta P_{change \text{ shell side}} = - \frac{1}{2} \frac{(\dot{m}_{shell}^t)^{t+\Delta t}}{\rho_{shell}^t A_{shell}^2} \left(f_{darcy} \frac{l}{d_h} + K \right)_{shell}^t \quad (8)$$

Do note that the density and the loss coefficients for the fluid within the shell at the last timestep ρ_{shell}^t and $\left(f_{darcy} \frac{l}{d_h} + K \right)_{shell}^t$ are used based on

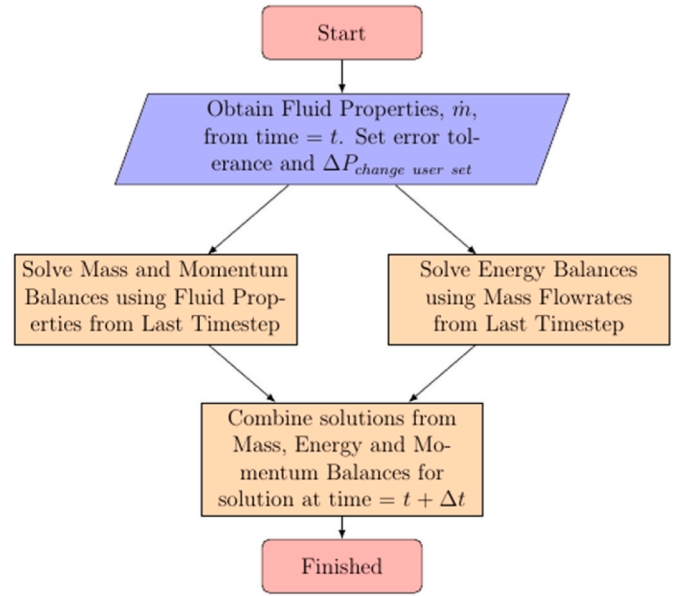


Fig. 4. Operator splitting method.

the fluid properties at the last timestep. Of course, in the Boussinesq approximation, differences in ρ_{shell}^t may be neglected, but the indices are added to illustrate the algorithm. Concurrently, the energy equations for the fluid in the shell side are also calculated using the fluid velocities (from the mass flowrates) calculated at the last timestep:

$$\rho_i c_p^t (T_x^t) A_{xs,x} \Delta x \frac{T_x^{t+\Delta t} - T_x^t}{\Delta t} = - \rho_i c_p^t (T_x^t) A_{XS,x} u^t (T_x^{t+\Delta t} - T_{x-\Delta x}^{t+\Delta t}) + \sum_j q_j^t A_j \quad (9)$$

Equation (9) can be calculated independently from equation (8). Hence both equations can be solved using separate parallel computation threads. The mass flowrates at the next timestep $\dot{m}_{shell}^{t+\Delta t}$ are then used to calculate $u^{t+\Delta t}$ for the energy equation at the next timestep. The temperature profile at the next timestep is then used to calculate the densities, heat capacities and dynamic viscosities required to compute $\left(f_{darcy} \frac{l}{d_h} + K \right)_{shell}^{t+\Delta t}$. This approach is used for the CIET Educational Simulator, but within TUAS the user has the flexibility to choose to solve equation (8) first, and use the $\dot{m}_{shell}^{t+\Delta t}$ to compute $u^{t+\Delta t}$ for use in Equation (9).

3.2.2. Iteration algorithms

To solve the mass and momentum balances, the Brent-Dekker algorithm was used as shown in Fig. 5 [34]:

In Fig. 5, the default bounds for $\dot{m}$ are between -10 kg/s to 10 kg/s. If the algorithm fails to find a root, the algorithm is restarted with a larger bound of $-10,000$ kg/s to $10,000$ kg/s. This is meant to save time as forced circulation flows, which are larger in magnitude than natural circulation flows, within CIET are on the order of 0.18 kg/s [21]. $\Delta P_{change \text{ guessed}}$ will be determined using guesses from within this bracket. The user set pressure change $\Delta P_{change \text{ user set}}$ is calculated using $\Delta P_{change \text{ branch}}$ in Equation (2). In the simplest case where one branch represents a closed loop, $\Delta P_{change \text{ branch}} = 0$ Pa.

In more complex cases where two or more branches are calculated in parallel, a nested iteration loop is used. First, Equation (1) is used to describe that the mass flowrate across a bundle of parallel branches is equal to the sum of flow through each branch. Second, the total mass flowrate in Equation (1) is set to 0 kg/s because there is no net flow through the parallel branches overall. Thirdly, the outer loop guesses a $\Delta P_{change \text{ branch}}$ which must be the same across all branches, and then

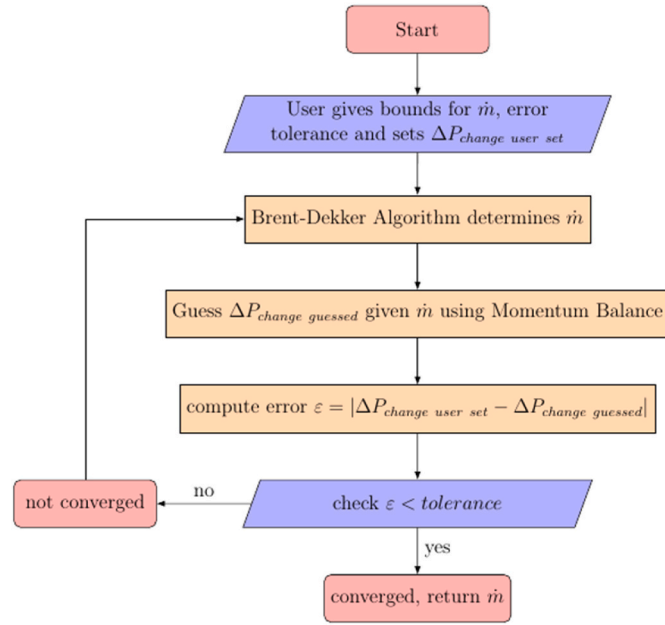


Fig. 5. Brent-Dekker algorithm for solving mass and momentum balances.

iteratively finds the mass flowrates across each branch using Fig. 5. This is the inner iteration loop. Fourthly, it sums them together and checks if it equals 0 kg/s.

The pressure change bounds for the outer loop are decided by setting the mass flowrate to 0 kg/s across each branch and then computing $\sum_i^n \rho_i g z_i + \sum_i^n \Delta P_{source,i}$ as shown in Equation (2) across each branch. $\sum_i^n \rho_i g z_i + \sum_i^n \Delta P_{source,i}$ represents the sum of natural circulation and forced circulation driving forces across each branch. The ΔP_{change} branch bounds are decided based on the highest and lowest $\sum_i^n \rho_i g z_i + \sum_i^n \Delta P_{source,i}$ across the branches.

For the semi-implicitly coupled energy balances, matrices were formed which required solvers to solve $\mathbf{A}\vec{x} = \vec{b}$. The TUAS Solver library used linear algebra package (LAPACK) libraries which relied on OpenBLAS [49] to solve these equations efficiently. These libraries have inbuilt solver algorithms which are not discussed in this work.

3.3. Code flow and structure for the Boussinesq Solver

The TUAS Solver was written as a medium level library to allow users to construct their own components and simulations with maximum flexibility but also abstract away code details such as matrix construction and solver iterations to simplify development. For basic usage, users will be able to use a set of pre-built components such as pipes, heaters and exchangers and connect them together to form a simulation. For more advanced usage, users will be able to connect control volumes (CVs) and boundary conditions (BCs) together to build their own components using the existing pre-built components as a template or reference. These are collectively known as heat transfer entities. When connecting heat transfer entities, the rate of heat transfer between the entities is computed depending on the type of interaction. This could be a mixture of conduction or advection interactions, or even a user specified heat flowrate. The user then initiates calculation of enthalpy at the next timestep in the timestep advancement stage. These are shown in Fig. 6 [34]:

As shown in Fig. 6, heat transfer rates are computed using the relevant heat transfer interactions. These are stored in a heat transfer rate (power) vector as Single CVs may interact with more than one BC or Single CV. At the timestep advancement stage, the enthalpies at the next timestep are computed and the power vectors are emptied.

Heat transfer components are made up of several CVs, and

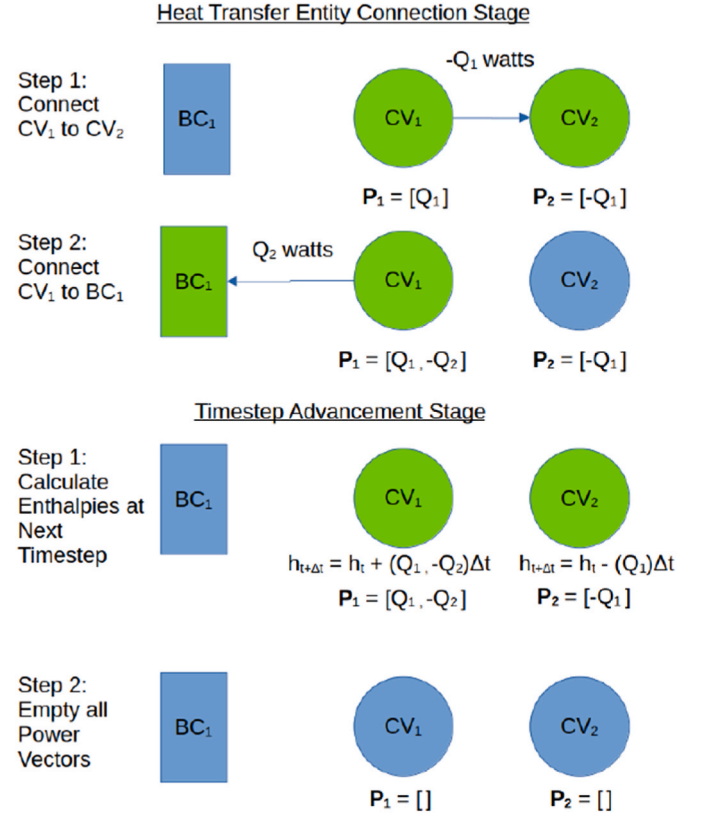


Fig. 6. Program flow for the tuas solver.

constructing a heat transfer component one CV at a time is inconvenient. To enable more user friendly construction of heat transfer components, one dimensional arrays of control volumes (ArrayCVs) are also programmed and provided for end users. Within ArrayCVs, a semi-implicit time coupling scheme is used, whereas axial coupling is done using explicit time coupling as shown in Fig. 7 [34]:

As shown in Fig. 7, axial coupling is done by explicit coupling of two Single CVs, thus the same steps and code in Fig. 6 were used. For a more complex component such as a STEH shown in Fig. 3, five ArrayCVs are coupled laterally in semi-implicit fashion using Equation (10):

$$P = H_{lateral} \left(T_{CV-self}^{t+\Delta t} - T_{lateral}^t \right) \quad (10)$$

For each lateral connection, a vector of thermal conductances and temperatures were obtained at the previous timestep during the

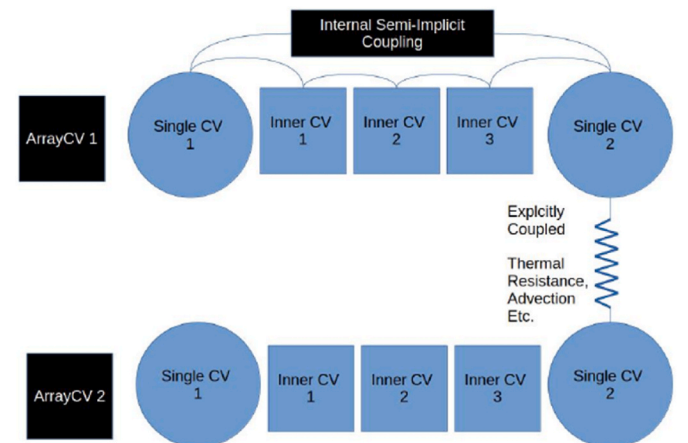


Fig. 7. Coupling schemes within and between ArrayCVs.

connection stage. During timestep advancement, the semi-implicit equations for temperature in the next timestep were then constructed in the form $\mathbf{A}\vec{x} = \vec{b}$, and solved within ArrayCVs using the LAPACK libraries.

SingleCVs, ArrayCVs and BCs are programmed using an object oriented programming (OOP) paradigm which abstracts away the need for the end user to think about energy balance equation and matrix construction during connection and timestep advancement. These objects contain functions (also known as methods) to aid the user in connection stage and timestep advancement stage. Moreover, programming with an OOP paradigm allows for code modularity and extensibility like open source codes such as Cyclus [50]. In practice, it is difficult to extend or use code that is difficult to read. Hence, the TUAS library modules are arranged in a manner of increasing abstraction and dependency as shown in Fig. 8:

These modules are programmed in the Rust as opposed to C++ for SAM using the MOOSE framework [43] and Fortran for the RELAP [51]. This is because Rust enabled software development allowing a similar execution speed to C++ and Fortran, but also with a lower risk of data races [52] and using a lower cognitive effort in contrast to C++ and Fortran [53]. This allowed for real-time simulation of thermal hydraulics within CIET [34]. Further documentation on the rationale for using Rust as well as the coding structure and algorithms were presented in as thesis and dissertation work [33,34]. A fuller exposition on the principles, code structure and quality control would require a full paper to discuss as was the case with Cyclus [50] and GeN-Foam [54]. This is therefore reserved for future work.

3.4. Component verification and validation

While preliminary verification and validation (V&V) of TUAS was done in previous work as a PhD Dissertation [34], the pre-built components in the DRACS V&V also need to be validated individually first. One of them is the SimpleShellAndTubeHeatExchanger component validated using Du's STHE. The general layout is shown in Fig. 3. However, both insulated and non-insulated pipes, the InsulatedFluidComponents and NonInsulatedFluidComponents respectively are used to construct the DRACS loop as well as the CIET Educational Simulator, and need to undergo V&V as individual components. To validate the NonInsulatedFluidComponents, De Wet's steady state

experimental data for the bare version of CIET's heater v2 is used [55] which was in Table 5.8 on page 158 of a dissertation [34]. The validation procedure for the NonInsulatedFluidComponents, essentially (optionally) heated pipes without insulation, can be visualized using Fig. 9, which is an example using data from the 8 kW steady state data:

NonInsulatedFluidComponents are used to model CIET's heater without the interior, and then subject to various inlet boundary conditions and heater powers. The outlet temperature at steady state is then compared to experimental data. In the actual experimental data, a static mixer and some piping is located between the heater outlet and the thermocouple measuring its temperature, labelled BT-12. However, the parasitic heat loss from them is assumed to be negligible (<0.5K). Hence, BT-12 is taken to be a proxy for the bulk temperature right at the heater exit modelled using NonInsulatedFluidComponents. These are used to model the flowmeters, DHX tube side and TCHX within the isolated DRACS loop verification case. The rest of the piping is InsulatedFluidComponent, which adds an insulation on top of Non-InsulatedFluidComponents. For the DRACS loop however, parasitic heat loss from them is set to zero as it is an idealized case as explained in Zou's work [36]. In this regard, V&V for the DRACS loop with zero parasitic heat loss is quite trivial.

However, for the CIET Educational Simulator, piping both forced and natural circulation are modelled with parasitic heat losses. The InsulatedFluidComponent objects are used to model these pipes. To ensure that these are programmed correctly, a verification procedure was carried out like Fig. 9, except that no heat was added to the piping, and that an extra layer of insulation was added on top of the piping. The outlet temperature (T_{out}) of an analytical solution, Equation (11), was compared against that of the InsulatedFluidComponent in TUAS. Equation (11), which can be derived from a log-mean temperature difference equation for heat exchangers is presented as:

$$T_{out} = (T_{in} - T_{ambient}) \exp\left(-\frac{UA}{\dot{m}c_p}\right) + T_{ambient} \quad (11)$$

From Equation (11), the overall thermal conductance (UA), increases with length. Hence, with increasing length, we expect more parasitic heat loss. Thus, T_{out} decreases with increasing pipe length. As parasitic heat losses may be small in individual pipes, it may be difficult to ascertain if the thermal resistances between the fluid, pipe and insulation were correctly programmed. Hence, parasitic heat losses were

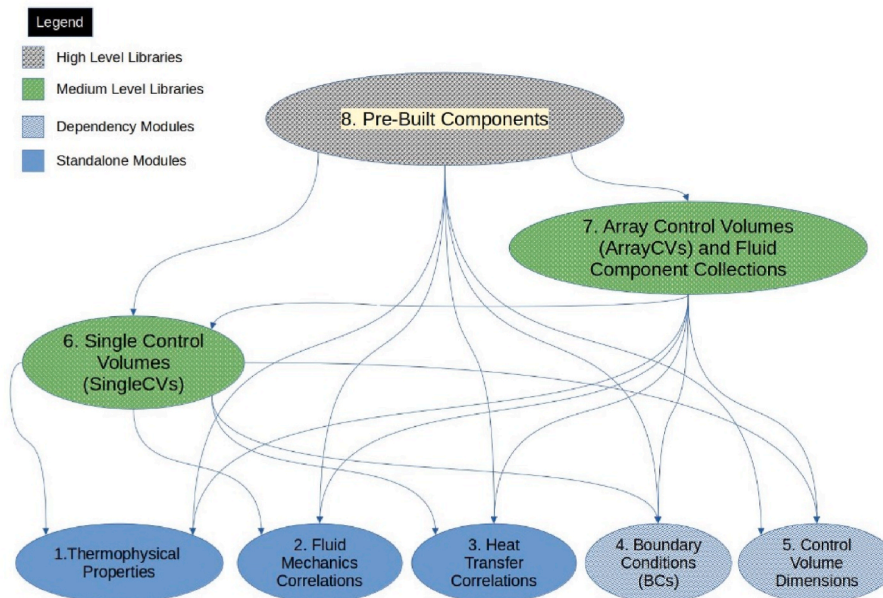


Fig. 8. Solver dependency graph.

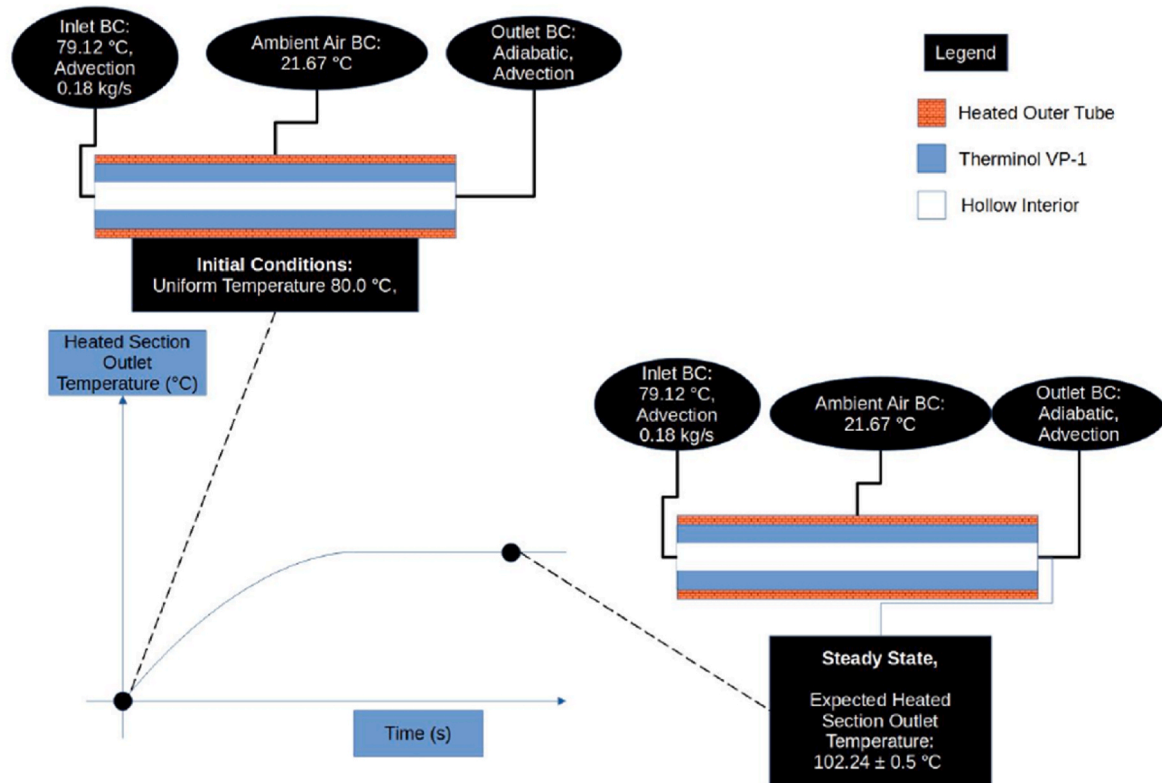


Fig. 9. Validation procedure for heated pipes [34].

artificially increased for the verification tests so that potential errors were easier to catch. For these tests, parameters for static mixer 41 (component 6 in Fig. 1) were used from the RELAP model found in Table A1 and Table A2 on page 111 and 112 of Zweibaum's PhD Dissertation [21]. However, parasitic heat losses were increased by reducing the pipe thickness and insulation thickness by 10. To further increase parasitic heat loss, the heat transfer coefficient to ambient was increased to $2000 \frac{W}{m^2K}$ and the Nusselt Number artificially increased to 4000. The inlet temperature was 100 °C and mass flowrate was 0.18 kg/s.

For these tests, the number of nodes used was 10 nodes per meter of pipe simulated plus two. Hence, for a 3 m pipe, 32 nodes were used, and for a 9 m pipe, 92 nodes were used. Timesteps used for the simulation were fixed at 0.1 s.

3.5. CIET educational simulator construction

As a step towards building real-time FHR simulators with 2D and/or 3D GUIs, the CIET educational simulator was constructed. This is primarily because CIET is a scaled simulant fluid IET facility of the FHR, but also because forced and natural circulation data for CIET is readily available in literature. Moreover the authors have had some prior experience in operating CIET experimentally, and therefore there was some intuition as to how operating CIET in real-time should feel like to an operator. The 2D GUI simulator was built using the Rust egui library [30] which shows that TUAS could integrate well with external GUI libraries, paving the way for future open-source 3D GUI simulators.

The CIET Educational Simulator takes advantage of Rust's versatility with multithreaded processing such that the mass flowrate equations, the energy equations and the graphical rendering were performed on separate computing threads so that the simulator could run in real-time from the perspective of user experience. Computational details are not discussed extensively in this work for the sake of brevity. However, engineering and simulation details for the CIET Educational Simulator

are still discussed because of its relevance to TUAS and transient simulation validation. Readers interested in computational details can freely refer to the TUAS Github repository for source code of the CIET Educational Simulator [37].

3.5.1. Forced and natural circulation modes

CIET can operate in forced and natural circulation mode. This capability is emulated in the CIET Educational Simulator. The DRACS loop is simulated using two parallel branches as described in Equation (2), whereas the DHX branch, Heater branch and CTAH branches are simulated using three parallel branches also described using Equation (2). The user is given the option to block flow in the CTAH and the DHX branch during simulation. This emulates valving off either branch in the CIET experiment for forced and natural circulation mode respectively.

CIET also has a flow diode that restricts flow through the DHX branch under forced circulation [21]. This is also programmed into the DHX branch in the CIET Educational Simulator. For simplicity, flow is only allowed if it goes from the top of the DHX branch to the bottom, it is completely blocked in the opposite direction. This is programmed by first solving for flow across the three parallel branches (three branch solution), provided the user does not block off the DHX branch. If flow from the DHX proceeds from top to the bottom, the mass flowrates across all three branches from the three branch solution is taken to be the mass flowrate at the next timestep. If flow in the DHX proceeds from bottom to top, the flow across the CTAH and Heater branches is solved (two branch solution) while flow in the DHX branch is set to 0 kg/s. This approach was taken in contrast to setting asymmetric flow resistances in the DHX branch because the latter approach tended to cause the Brent-Dekker iteration algorithm to converge slowly, thus increasing solution time. The mass flowrates for the CTAH and DHX branch from two branch solution is then taken to be the mass flowrate for the next timestep. Now, solving for the DRACS loop mass flowrates, three branch solution, and two branch solution serially proved to be expensive in terms of computation time. Therefore, the DRACS mass flowrate, two

branch solution and three branch solution are calculated in parallel computation threads. As before, the choice of which solution is used for the next timestep is determined by the flow across the DHX.

For the CTAH pump, simplifications are made so that the user sets the manually sets the dynamic pressure drop across the pump. Simulation of pump curves, pump head, and the variable frequency drives in the CTAH pump are left to future work.

3.5.2. Proportional, Integral and Derivative (PID) controllers

For the CIET Educational Simulator v1, filtered PID controllers were used to allow the user to control the CTAH and the TCHX coolers. In the CTAH and TCHX models, simplifications were made so that the heat transfer coefficient was directly controlled by the controllers. Modelling fan speeds and their empirical relationships with the heat transfer coefficients is left to future work. These controllers were based on another open source controller library developed for previous work [34]. The transfer function for the said filtered PID controller for the TCHX was:

$$G(s) = K_c + \frac{K_c}{\tau_i s} + K_c \frac{\tau_d s}{\alpha \tau_d s + 1} \quad (12)$$

$$\text{Where } G(s) = \frac{\bar{O}(s)}{\bar{E}(s)}, \bar{O}(s) = \frac{h_{TCHX} - 40 \left(\frac{W}{m^2 K} \right)}{40 \left(\frac{W}{m^2 K} \right)} \text{ and } \bar{E}(s) = \frac{T_{TCHX, \text{outlet}} - T_{TCHX, \text{outlet sp}}}{80 K}.$$

For the TCHX, $K_c = 70.75$, $\tau_i = 0.2s$, $\tau_d = 1s$, and $\alpha = 1$. The lower bound of $h_{TCHX} = 5 \frac{W}{m^2 K}$ was provided to ensure that the TCHX would not inadvertently provide heat to the fluid.

For the CTAH, Equation (12) is also used, except that $(s) = \frac{\bar{O}(s)}{\bar{E}(s)}$,

$$\bar{O}(s) = \frac{h_{CTAH} - 40 \left(\frac{W}{m^2 K} \right)}{40 \left(\frac{W}{m^2 K} \right)} \text{ and } \bar{E}(s) = \frac{T_{CTAH, \text{outlet}} - T_{CTAH, \text{outlet sp}}}{80 K}. \text{ For the CTAH, } K_c =$$

70.75 , $\tau_i = 0.02s$, $\tau_d = 0.01s$, and $\alpha = 1$. The lower bound of $h_{CTAH} = 5 \frac{W}{m^2 K}$ was provided to ensure that the CTAH would not inadvertently provide heat to the fluid. These controller settings allowed for relatively responsive and stable control of the CTAH and TCHX.

In early testing, the PID controllers became sluggish in response when the outlet temperature deviated from the set points for excessive periods of time. This is because the integral term of the PID controller kept growing to unrealistically large values in that time and took excessively long to reset. To increase the responsiveness of the PID controllers for convenient user experience, the integral term was set to zero if the heat transfer coefficient fell outside the arbitrary bounds of $5 \frac{W}{m^2 K}$ to $10,000 \frac{W}{m^2 K}$.

3.5.3. Component modelling and nodalisation as well as time and spatial discretization

For the GUI simulator, the spatial discretization was determined using the SAM nodalisation scheme found in page 23 of Zou's SAM publication [18]. For this nodalisation scheme, time steps are set at 0.1s by default as this was the timestep required to simulate forced circulation without numerical instability. The user is also given the option to reduce the time step during simulator runs to put the simulator in slow motion mode.

One difference between the CIET Educational Simulator and SAM is the addition of two adiabatic mixing nodes for where the three branches in the primary loop meet as seen in Fig. 1. This is between pipe 4, 5a and 5b, as well as pipe 17a, 17b and 18. These nodes were added as it was programmatically easier to use one control volume to connect the three branches and help them mix. The thermal inertia of these nodes was arbitrarily set by programming them as 3.84 cm diameter spheres of Dowtherm A. This is meant to be small enough to minimize the increase of thermal inertia in the simulator but large enough so as not to introduce numerical instability given the 0.1s timestep. These would not

impact steady state temperatures but would impact temperatures observed during transient. This setup is taken as a baseline which then can be improved upon in future work where more V&V studies can be performed.

3.5.4. Initial calibration, verification and validation

The CIET educational simulator as it stands in this work (v0.0.6 of TUAS) is meant to showcase TUAS's ability to function well as a tool for public outreach and education. However, with proper steps taken, the CIET educational simulator can also function as a systems level simulation of CIET in forced and natural circulation mode used for transient studies and research. For this purpose, V&V studies must be performed, and a limited V&V study for forced circulation was performed to showcase the capabilities of the CIET Educational Simulator. This initial calibration and transient validation effort was also performed to showcase TUAS's ability to reproduce some forced circulation transient experimental data from CIET despite using a first order upwind scheme. However, full V&V of the CIET educational simulator is reserved for future work. For these tests, experimental data for heater power, as well as inlet and outlet temperatures on Fig 2.21 on page 47 of Zweibaum's dissertation [21] were used. The GraphReader software was used [56] the relevant experimental data and convert them into csv data.

For initial calibration, the insulation thicknesses were reduced in all branches shown in Fig. 1. This is to ensure that parasitic heat losses matched that of the experiment for forced circulation like what was done for the RELAP model [21]. The heater Nusselt Number from the Therminol VP-1 to the outer pipe was also increased by 1.6 times as part of interim parasitic heat loss calibration. A summary of important parasitic heat loss calibration parameters is shown in Table 1:

Besides these adjustments, interim calibration was performed for the coupled natural circulation loop. This involved the heater branch and DHX branch. This is also why the heater branch components have different insulation thicknesses from the CTAH branch. However, validation of coupled natural circulation, whether steady state or transient, are outside the scope of this work. Since calibration is still ongoing and subject to change, calibration values of the DHX branch and DRACS loop are not reported. Interested readers can view the source code in TUAS v0.0.6 on Github for more details on the calibration [37]. As mentioned previously, thermal inertia calibration is also not performed, and therefore, this is also left to future work.

4. Results and discussion

4.1. Isolated DRACS natural circulation loop code to code verification

Using the algorithms and solvers as described in part 3.2, a simulation of the DRACS loop was constructed, and satisfactory results for code to code verification with SAM were obtained as of

Table 1
Summary of parasitic Heat loss calibration parameters for forced circulation.

Components	Fiberglass Insulation Thickness (cm)	Heat Transfer Coefficient to Ambient ($\frac{W}{m^2 K}$)	Remarks
1a, 2a, 2, 3, 4	0.24	20.0	Heater Branch after Heater
1b, 18	0.15	20.0	Heater Branch before Heater
5b, 6, 6a	0.5	20.0	From Heater Branch to CTAH
8a, 8, 9, 10, 11, 12	1.00	20.0	From CTAH to CTAH Pump
13, 14, 15, 16, 17a	1.00	20.0	From CTAH pump to Heater Branch
1	5.08	20.0	Heater

tuas_boussinesq_solver v0.0.2. These simulations also double as regression tests that must be passed when TUAS is updated to newer versions. To pass these tests, the natural circulation mass flowrates generated by newer versions must agree ideally to within 0.44 % of the mass flowrates in v0.0.2. This 0.44 % was the maximum percentage error between SAM and the analytical solution in literature [18]. The setup for these regression tests were based largely on the schematics in Fig. 1, except that the vertical portion NDHX or TCHX (35b-1 and 35b-2) was combined into one component. The nodalisation of the TCHX into 35b-1 and 35b-2 was done only in the coupled natural circulation loop for SAM as part of model calibration [18]. Pressure drop correlations and dimensions were obtained mostly from the SAM model, but the TCHX dimensions were not found in the published SAM model to the best of the author's knowledge. Therefore, the parameters from the RELAP model were used [21]. Moreover, where there were insignificant discrepancies in pipe dimensions between RELAP and SAM, the RELAP parameters were used due to legacy reasons. A full dossier of the parameters meant for constructing the isolated DRACS can be found in supplementary data on request. The TCHX, DHX tube side and flowmeter were modelled using NonInsulatedFluidComponent objects. These were validated using CIET Heater v2 experimental data shown in Fig. 10:

Fig. 10 shows that the NonInsulatedFluidComponent objects are validated to within 0.5K thermocouple measurement uncertainty. The remaining objects have been programmed using adiabatic piping. In such a case, inlet temperature should equal outlet temperature for any flowrate. In this case, V&V is trivial and not presented here. In the isolated DRACS loop verification case, no parasitic heat loss was assumed throughout the loop. Therefore, many components were modelled using adiabatic piping. As V&V for adiabatic piping in the DRACS loop is trivial, it is not presented in this work.

For the isolated DRACS loop code to code verification with SAM, three datasets were generated based on the SAM isolated DRACS loop simulations [36]. Each dataset corresponded to a fixed TCHX outlet temperature in natural circulation. To ensure that the TCHX outlet temperature was fixed, Proportional, Integral and Derivative (PID) controllers were used like what was done in the FHR advanced natural circulation (FANCY) code [57]. For these simulated natural convection loops, the same PID controllers were used, and $K_c = 1.75$, $\tau_I = 1.75s$, $\tau_d = 1s$, and $\alpha = 1$. The lower bound of $h_{TCHX} = 5 \frac{W}{m^2K}$ was provided to ensure that the TCHX would not inadvertently provide heat to the fluid. This allowed the simulation to remain somewhat physically realistic as this would be roughly equivalent to turning off the fans of the TCHX. The simulation was run for 3800s of simulated time at timesteps of 0.5s to allow the system to reach steady state. The results are presented in Table 2, Tables 3 and 4:

The results shown in Table 2, Tables 3 and 4 show that the TUAS solver generally agrees well with the SAM results. However, there is still a 1–2.6 % deviation between the TUAS library and SAM. This value is significant compared to the maximum percentage error between SAM and the analytical solution of 0.44 % [18]. This large error, compared to

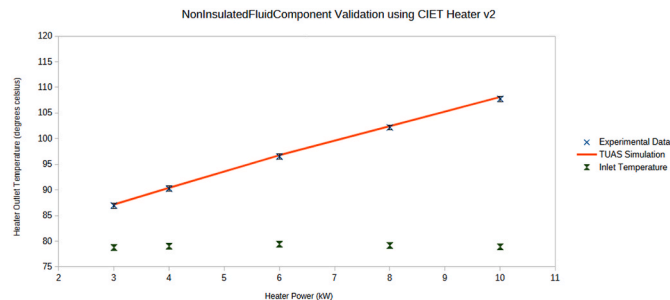


Fig. 10. Validation for Non-Insulated Heated Piping Components (Non-InsulatedFluidComponent), Experimental Data marked by Blue Crosses.

Table 2

Isolated DRACS dataset A, TCHX outlet temperature 46 ° C.

Input Power (Watts)	$\dot{m}_{TUAS}$ (kg/s)	$\dot{m}_{SAM}$ (kg/s)	$\frac{\dot{m}_{TUAS} - \dot{m}_{SAM}}{\dot{m}_{SAM}} \times 100\%$
931.8	3.5444E-02	3.5019E-02	1.21
1088.3	3.7794E-02	3.7275E-02	1.39
1338.4	4.1135E-02	4.0598E-02	1.32
1470.6	4.2735E-02	4.2127E-02	1.44
1699.9	4.5315E-02	4.4674E-02	1.43
1876.5	4.7145E-02	4.6411E-02	1.58
2137.0	4.9653E-02	4.8867E-02	1.61

0.44 %, was likely due to discrepancies in flow area and hydraulic diameter of the TCHX between SAM and that used in this work, where these parameters were taken from the RELAP model [21]. If there was indeed a discrepancy between flow area and hydraulic diameter, we should observe that the systematic error in mass flowrate grows with increasing DHX power due as the differences in flow area and hydraulic diameter result in an effective difference in loop pressure drop correlation. To understand the cause of this difference, the following equations for the same component with slightly different form and frictional loss correlations can be considered.

$$\Delta P_{loss,1} = \frac{1}{2} \left(f_{darcy} \frac{l}{d_h} + K \right)_1 \rho u_1^2 \quad (13)$$

$$\Delta P_{loss,2} = \frac{1}{2} \left(f_{darcy} \frac{l}{d_h} + K \right)_2 \rho u_2^2 \quad (14)$$

Given the same dynamic pressure $\frac{1}{2} \rho u^2$, the discrepancy in loop pressure drop scales as $\frac{1}{2} \rho u^2$:

$$\Delta P_{loss,1} - \Delta P_{loss,2} = \frac{1}{2} \rho u^2 \left[\left(f_{darcy} \frac{l}{d_h} + K \right)_1 - \left(f_{darcy} \frac{l}{d_h} + K \right)_2 \right] \quad (15)$$

Suppose that we plot equation (15) out graphically, we would observe that the discrepancies in mass flowrate scale with increasing driving force. This driving force is the density difference between hot and cold leg arising from temperature difference. To calculate this temperature difference, for each dataset the cold leg temperature is roughly constant, and the hot leg temperature increases with increasing power. Therefore, with increasing power at a constant TCHX outlet temperature, there should be an observed increase in driving force and therefore mass flowrate error. This is indeed the trend shown in the across all datasets in Table 2, Tables 3 and 4.

In Table 2, Tables 3 and 4, the error magnitudes grow with increasing power in the same dataset. Moreover, there is a systematic over-prediction of mass flowrates for the TUAS library DRACS loop as compared to the SAM DRACS loop. Both observations are consistent with the notion that there were inconsistencies in hydraulic diameter and flow area for the TCHX. Verifying if this was indeed the case is difficult because TCHX parameters from the SAM model were not published in Zou's publication. The most obvious source of potential discrepancy was that TUAS used an first-order upwind scheme whilst

Table 3

Isolated DRACS dataset B, TCHX outlet temperature 35 ° C.

Input Power (Watts)	$\dot{m}_{TUAS}$ (kg/s)	$\dot{m}_{SAM}$ (kg/s)	$\frac{\dot{m}_{TUAS} - \dot{m}_{SAM}}{\dot{m}_{SAM}} \times 100\%$
454.4	2.4663E-02	2.4343E-02	1.31
766.2	3.1149E-02	3.0558E-02	1.93
1004.4	3.5051E-02	3.4364E-02	2.00
1211.2	3.7992E-02	3.7117E-02	2.36
1409.0	4.0523E-02	3.9599E-02	2.33
1607.4	4.2844E-02	4.1851E-02	2.37
1804.6	4.4977E-02	4.3913E-02	2.42
2004.9	4.6995E-02	4.5834E-02	2.53
2211.0	4.8940E-02	4.7704E-02	2.59

Table 4

Isolated DRACS dataset C, TCHX outlet temperature 40 ° C.

Input Power (Watts)	$\dot{m}_{TUAS}$ (kg/s)	$\dot{m}_{SAM}$ (kg/s)	$\frac{\dot{m}_{TUAS} - \dot{m}_{SAM}}{\dot{m}_{SAM}} \times 100\%$
582.6	2.8358E-02	2.8033E-02	1.16
785.9	3.2272E-02	3.1807E-02	1.46
971.4	3.5319E-02	3.4690E-02	1.81
1185.2	3.8408E-02	3.7765E-02	1.70
1369.1	4.0793E-02	4.0100E-02	1.73
1584.1	4.3334E-02	4.2499E-02	1.96
1763.7	4.5292E-02	4.4448E-02	1.90
1970.0	4.7387E-02	4.6455E-02	2.01
2177.0	4.9350E-02	4.8315E-02	2.14

SAM used a second-order scheme. This would impact the conduction term and therefore temperature profile along the pipe. The result is a different natural circulation driving force and a differing natural circulation flowrate. To mitigate numerical diffusion, a finer mesh could be used and the results compared. To investigate if the lack of mesh refinement caused significant deviations in flowrate, all cases for Dataset C were repeated using twice the number of nodes in all components compared to the SAM nodalisation and a timestep of 0.25s (mrf-a), and then a further test where five times the number of nodes were used with a timestep of 0.1s (mrf-b). Mesh refinements at 10 and 20 times the number of nodes for SAM with even shorter timesteps were attempted to check if the mass flowrates converged. However, these tests were not completed even after several hours and were slower than real-time and hence were skipped.

Based on these results in Table 5, significant improvements in the mass flowrates were observed across all the test cases. These 18 tests, each simulating 3800s of simulation time took approximately 600s to complete on a i7-10875H AFTERSHOCK Gaming Laptop running Arch Linux. Despite the mesh refinements, TUAS is still unable to reproduce the results of the SAM model to within 0.44 %. The lack of published data for the SAM model for the TCHX also makes it difficult to reproduce the results for the isolated DRACS loop from SAM exactly, or at least to within 0.44 %. Again, further mesh refinements could be attempted, but in those cases, TUAS loses the potential for real-time simulation capability, which is a key performance metric for real-time GUI educational simulators. In this situation, the coarse mesh solution still has value as it can be run faster and is reasonably accurate. To retain TUAS's potential for real-time simulation, we still need to consider if an error of 2.6 % for the coarse mesh results is too large.

To establish if the 2.6 % error is significant, it is useful to consider typical error magnitudes in code-to-code verification for natural circulation. For this, one can consider the simulation of natural circulation using the codes RELAP5 and FANCY. Zweibaum reported that RELAP5 agreed with CIET experimental data (natural circulation mass flowrates) to within 1 % whilst the FANCY natural circulation code agreed with the same experimental data to within 3 % [20]. Based on this information, we can perform a crude estimate that the discrepancy between RELAP

and FANCY is around 2–4 %. Since the 2.6 % error between the TUAS library solver and SAM is within the range of 2–4 %, we can consider that a 2.6 % discrepancy is typical of that found in literature. Based on this, it can be concluded that even with errors arising from first order schemes, the TUAS library can be considered sufficiently verified.

In terms of runtime, the TUAS library solver showed the capability of real-time simulation where the solver execution time is less than the simulated time. This was shown as all 25 tests for Case A, B and C were able to be completed within roughly 220s to complete for a 3800s simulation. This was done using parallel computing on an Arch Linux laptop with a i7-10875H 8 core 16 thread CPU used in previous work [34] which represents a typical gaming laptop computer in 2021. For each natural circulation test case, only one computing thread was assigned to each test case, which meant test cases were run serially. Thus, assuming each test case with 3800s simulation time took 220s to run, each test case was able to run the isolated DRACS natural circulation loop with PID controllers faster than real-time. On the Linux Mint i5-13500H laptop, the same tests took about 98s to run. Therefore, even though the Boussinesq solver cannot account for multiphase flow and salt freezing [43], it is valuable for open-source real-time simulation of natural circulation loops. Moreover, computer hardware requirements for real-time simulation of this Boussinesq solver are readily accessible in the market. Hence, users with the relevant skills and hardware can potentially run real-time natural circulation cases on their home computers. Of course, for true real-time simulation, TUAS will need to produce results for transient natural circulation in real-time. To verify if TUAS has this capability, a transient natural circulation validation study will need to be set up. This is left for future work. Nevertheless, the fact that the runtimes were less than the simulation time gives confidence that real-time simulation may be possible in future using TUAS.

Given the satisfactory run time performance of the TUAS library solver, and the good agreement between the TUAS library solver and SAM for the isolated DRACS models, the isolated DRACS natural circulation code to code verification effort is considered successful. This implies that this TUAS library solver could potentially simulate other single phase natural circulation loops, including the coupled loop natural circulation in CIET and other molten salt natural circulation loops.

4.2. Shell and tube heat exchanger (STHE) code validation

Similar to the code-to-code verification effort, the STHE code validation effort is reasonably successful in reproducing Du's Nu_s as shown in Fig. 11:

In Du's experiment, raw experimental data of the STHE's temperatures underwent a data reduction technique which converted all of them into Nusselt Numbers plotted as " Nu_{expt} " in Fig. 11 [40]. These Nusselt numbers were then used to form an empirical correlation which was Equation (7). Fig. 11 shows that Nu_s for the simulation datasets A, B and C agree with experimental data (Nu_{expt}) within error bars of 8 %. This 8 % was chosen because the discrepancy between experimental data and empirical correlation in Equation (7) was also at most 8 % [40]. Therefore, an 8 % error margin should be reasonable. These results were produced by constructing the STHE simulation, running them using test parameters provided by Du et al., and then performing a similar data reduction technique employed by Du et al. In this data reduction technique, the overall heat transfer coefficient (U) was first calculated before it was then used, along with estimated Re and Pr to calculate the Nu_s .

To construct the STHE simulations which model the STHE in Du's work, the STHE parameters from Du's publication [40]. Based on these parameters, a STHE simulation with control volume setup similar to Fig. 3, except that 14 axial control volumes were used. Simulations were run for roughly 1500s, using timesteps of about 0.035s. The inlet temperatures of both shell and tube side were set using an inlet boundary condition whilst the outlet boundary condition in both shell and tube was adiabatic. The results of the simulations are presented in Table 6, Tables 7 and 8:

Table 5

Results of mesh refinement tests for dataset C.

Input Power (Watts)	$\dot{m}_{TUAS \text{ mrf-a}}$ (kg/s)	$\dot{m}_{TUAS \text{ mrf-b}}$ (kg/s)	$\frac{\dot{m}_{TUAS \text{ mrf-b}} - \dot{m}_{SAM}}{\dot{m}_{SAM}} \times 100\%$
582.6	2.8273E-2	2.8220E-2	0.667
785.9	3.2172E-2	3.2113E-2	0.962
971.4	3.5213E-2	3.5149E-2	1.32
1185.2	3.8294E-2	3.8226E-2	1.22
1369.1	4.0601E-2	4.0601E-2	1.25
1584.1	4.3208E-2	4.3133E-2	1.49
1763.7	4.5161E-2	4.5082E-2	1.43
1970.0	4.7251E-2	4.7170E-2	1.54
2177.0	4.9210E-2	4.9125E-2	1.68

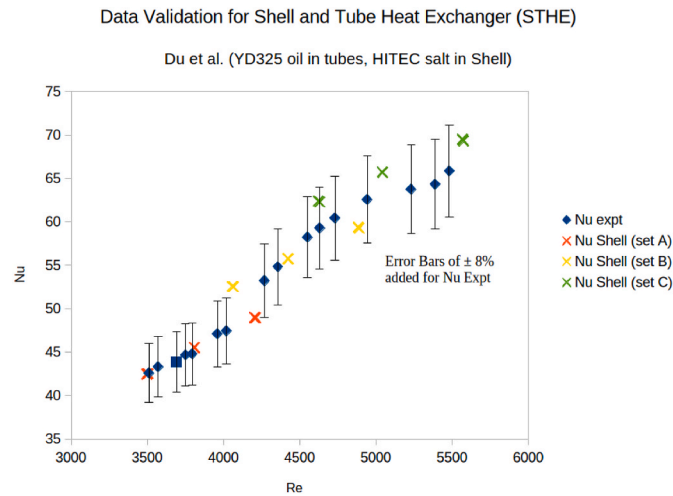


Fig. 11. Validation results from Du's molten salt Heat exchanger [40].

Table 6, Tables 7 and 8 restrict the inlet shell and tube temperatures as well as volumetric flowrates based on Du et al.'s [40] given parameter range except for the salt flowrate in dataset C. In dataset C, salt flowrate is slightly more than the Du et al.'s upper limit of $16.23 \frac{m^3}{h}$ [40].

To check if the resulting data is reasonably correct, we check the shell and tube outlet temperatures and U against Du's parameters [40]. The resulting shell outlet temperatures (that of HITEC salt) are from 203.27°C in set A1 to 223.94°C in set C7. The shell outlet temperature Set A1 is slightly lower than the lower bound temperature of 204.06°C , just under two times the $\pm 0.5\text{ K}$ uncertainty of thermocouples in Du's publication [40]. Given the lack of the published data used to produce Du's experimental dataset, it is judged acceptable to tolerate a this degree of discrepancy. The oil outlet temperatures range from 89.02°C in set A1 to 111.15°C in set C7. This is within the $\pm 0.5\text{ K}$ thermocouple uncertainty in Du's publication [40]. Thus, we can conclude that the outlet shell and tube temperatures were reproduced reasonably to the range provided in Du et al.'s publication [40].

For overall heat transfer coefficient, U , Du's provided range was from 548.45 to $776.85 \pm 4.10 \frac{W}{m^2 K}$ [40]. Within this uncertainty, the upper limit of U is $808.7 \frac{W}{m^2 K}$. All datasets, except C7, fall within this range. Set C7 in Table 8 is above the $776.85 \frac{W}{m^2 K}$ to within 8.6% . The exact cause of deviation cannot be determined conclusively because the original conditions to reproduce Du's experimental data are not available. Other than this outlier, we can conclude that U agrees with Du's provided range.

To investigate if the increased U in set C7 would adversely impact the ability of the simulation to reproduce the Re_s and Nu_s plot, set C7 was still included in the data reduction process and plotted in Fig. 11. From Fig. 11, set C7 can be seen as one of the data points at the rightmost part of the data plot. The Nu_s for set C7 follows the trend where increasing Re_s results in increased Nu_s . Moreover, the Nu_s of set C7 also falls within the 8% error bar of the rightmost experimental datapoint by Du et al.

Table 6

STHE validation dataset A, salt flowrate = $12.63 \frac{m^3}{h}$, oil flowrate = $15.635 \frac{m^3}{h}$

Simulation no.	Salt Inlet Temperature ($^\circ\text{C}$)	Oil Inlet Temperature ($^\circ\text{C}$)	Salt Outlet Temperature ($^\circ\text{C}$)	Oil Outlet Temperature ($^\circ\text{C}$)	$U \left(\frac{W}{m^2 K} \right)$
A1	214.93	74.49	203.27	89.02	591
A2	224.93	74.49	211.91	90.64	616
A3	236.91	74.49	222.23	92.57	643
A4	236.91	80.41	222.38	98.21	662
A5	214.93	90.41	203.87	103.98	636
A6	224.93	80.49	212.08	96.33	635
A7	236.91	90.49	222.74	107.68	693

Therefore, one can conclude that the higher U did not adversely affect the Nu_s predicted from set C7. Due to this anomalous datapoint, the Reynolds, Prandtl and Nusselt numbers calculated for set C7 is presented in Table 9 [40].

In Table 9, $Re_{f,shell}$ is Re_s , only that it was denoted (f) to show that bulk fluid properties were used to calculate it. This is to distinguish it from dimensionless numbers based on fluid properties calculated at wall temperatures, denoted (w), as it is for $Pr_{w,shell}$. Nu , Re and Pr for set A and B are in the supplementary data.

Now, to check the validity of the results in Re , Pr and Nu , the values in Table 9 as well as dataset A and B are compared to the given ranges provided by Du et al. For the molten salt Re_s , Du et al. provided Re_s in the range 3514 – $5482 \pm 2.94\%$ and Nu_s in the range 42.60 – $65.95 \pm 4.61\%$ [40]. Based on dataset A and Table 9, Re_s ranged from 3496 in Set A1 to 5575 in Set C7. For Re_s of Set C7, it is just less than 2% higher than the Re_s upper bound of 5482 , and therefore within the uncertainty bound of Re provided by Du.

For the molten salt $Pr_{f,shell}$, Du et al. mentioned that the $Pr_{f,shell}$ range is $19.82 < Pr_{f,shell} < 24.03$ [40], uncertainty bounds were not explicitly mentioned in the publication to the best of the author's knowledge. Therefore, the uncertainty bounds have to be estimated. Given the typical uncertainties of 2.94 and 4.61% for Re_s and Nu_s , $Pr_{f,shell}$ could be crudely estimated to have an uncertainty of ~ 1 – 2% (roughly the difference between the uncertainties of Re_s and Nu_s), a similar order of magnitude for uncertainties in Re and Nu . For datasets A, B and C, the $19.99 < Pr_{f,shell} < 24.09$. The high bound Pr is from dataset A5 from Table 2. This is at most 0.2% higher than $Pr_{f,shell}$ of 24.03 . In this case, the discrepancy is about one order of magnitude smaller than the expected uncertainty of $Pr_{f,shell}$. Du et al. also provided $0.4273 < \frac{Pr_{f,shell}}{Pr_{w,shell}} < 0.5646$ [40]. However, $0.5275 < \frac{Pr_{f,shell}}{Pr_{w,shell}} < 0.6357$ was obtained in this work, which is significantly different from the range provided for the wall Prandtl number correction factor. This is perhaps due to differences in the way $Pr_{w,shell}$ is estimated. In datasets A, B and Table 9, $Pr_{w,shell}$ is estimated at 37.89 . This is the $Pr_{w,shell}$ obtained at 440 K , the lower bound temperatures for thermophysical property correlations of HITEC [41]. In Du's work however, the correlations may have been used beyond their suitable temperature bounds to obtain a different $Pr_{w,shell}$, thus the $\frac{Pr_{f,shell}}{Pr_{w,shell}}$ correction factor may be different as compared to that seen in this work. To estimate the impact of this correction factor, one can note that the purpose of this correction factor is to more accurately estimate Nu_s . Hence, to determine if these differences matter, one must consider Nu_s . Based on Fig. 11, this discrepancy did not matter with regard to the accuracy of obtaining Nu_s . Hence, these discrepancies of how $Pr_{w,shell}$ was calculated may not matter significantly for the purposes of this work.

The molten salt Nu_s , ranged from 42.47 in Set A1 to 69.32 in Set C7. While Set A1 is within the 4.61% uncertainty of Nu_s , the Nu_s in set C7 is about 5.2% higher than the upper limit of 65.95 . This is slightly outside the 4.61% uncertainty range provided by Du. While this may be the case, the discrepancy may still be acceptable given that the Nusselt numbers fall within the 8% correlation uncertainty in Equation (7) [40]

Table 7STHE validation dataset B, salt flowrate = $14.63 \frac{m^3}{h}$, oil flowrate = $15.635 \frac{m^3}{h}$

Simulation no.	Salt Inlet Temperature (°C)	Oil Inlet Temperature (°C)	Salt Outlet Temperature (°C)	Oil Outlet Temperature (°C)	$U \left(\frac{W}{m^2 K} \right)$
B1	214.93	74.49	203.66	90.74	665
B2	224.93	74.49	212.43	92.41	688
B3	236.91	74.49	222.91	94.44	713
B4	236.91	80.41	223.01	100.1	737
B5	214.93	90.41	204.18	105.66	720
B6	224.93	80.49	212.57	98.1	711
B7	236.91	90.49	223.31	109.56	775

Table 8STHE validation dataset C, salt flowrate = $16.63 \frac{m^3}{h}$, oil flowrate = $15.635 \frac{m^3}{h}$

Simulation no.	Salt Inlet Temperature (°C)	Oil Inlet Temperature (°C)	Salt Outlet Temperature (°C)	Oil Outlet Temperature (°C)	$U \left(\frac{W}{m^2 K} \right)$
C1	214.93	74.49	204.14	92.15	726
C2	224.93	74.49	213.02	93.88	748
C3	236.91	74.49	223.6	96.02	773
C4	236.91	80.41	223.68	101.68	800
C5	214.93	90.41	204.6	107.06	791
C6	224.93	80.49	213.13	99.59	774
C7	236.91	90.49	223.94	111.15	843

Table 9STHE Re Pr and Nu for set C Salt Flowrate = $16.63 \frac{m^3}{h}$, Oil Flowrate = $15.635 \frac{m^3}{h}$

Simulation No.	$Re_{f,shell}$	$Pr_{f,shell}$	$Pr_{w,shell}$	Nu_s
C1	4623	24.05	37.89	62.40
C2	5041	22.06	37.89	65.75
C3	5567	19.99	37.89	69.54
C4	5568	19.99	37.89	69.45
C5	4633	24.00	37.89	62.31
C6	5043	22.05	37.89	65.69
C7	5575	19.97	37.89	69.32

shown as error bars in Fig. 11. Moreover, Du's original set of experimental conditions were not available, so attempting to reproduce Nu_s to within 4.61 % is not possible.

On the matter of computation time, the STHE simulation was proven to be able to run in real-time. On a Linux Mint laptop using a i5-13500H, using a single thread for each simulation, the simulation finished in just under 18s. And on an Arch Linux laptop with a i7-10875H, it took just under 34s. Since a 1500s long simulation took at most 34s, the potential for real-time simulation capability was demonstrated on consumer laptops for the STHE.

Despite limitations in the data available and the discrepancies described earlier, the STHE simulation programmed using the TUAS library solver was still able to predict Nu_s and Re_s to within 8 % of experimental data. This means that even with errors arising from first order schemes, the validation effort for the TUAS library solver using the Du's STHE data was successful.

4.3. CIET educational simulator

Given the success of the steady state V&V of TUAS, V&V for transient cases still had to be performed. Moreover, TUAS was built for the purpose of building 2D and 3D simulators for public outreach and education. To address these twin problems, the CIET Educational Simulator was constructed, and a simple validation case study was run for the Heater within the same simulator. For this validation study of this interim simulator, the coupled natural circulation loop, including the STHE code was not yet validated using experimental data. Hence, the

natural circulation loop in v0.0.6 should not be taken to be accurate yet, but only serves to demonstrate the future capability of the CIET Educational Simulator to represent such loops using a GUI. V&V of the natural circulation loop is relegated to future work. Nevertheless, this partial validation test shows the ease with which transient V&V tests can be performed using the simulator as prior coding knowledge in Rust is not strictly required. Full V&V studies for the CIET Educational Simulator can be performed as well but these studies are relegated to future work.

4.3.1. User interface features and cross platform capability

The CIET Educational Simulator contains a full diagram of CIET. This interface was inspired by a free and open-source chemical process simulator known as DWSim [58], and the artwork for heat exchangers, heaters, pumps and coolers used in this first version was taken from screenshots of the same open-source DWSim [58]. The user can visualize temperature profiles across the loop as the components change colours according to their fluid temperatures. This is shown in Fig. 12:

As shown in Fig. 12, the user is also able to change the set point of the CTAH, heater power, and pressure drop across the CTAH pump with the use of sliders on the main page. A colour legend is present on the right to help the user roughly gauge the temperature across each pipe component. However, for precise temperatures, only the inlet and outlet temperatures for the heater and CTAH are displayed to avoid overloading the user with information. Mass Flowrates across the DHX and CTAH branches are displayed using text on the flowmeters FM40 and FM20 as shown in Fig. 12. This provides the user with mass flowrate information useful in validation studies.

The CIET Educational Simulator is also able to run in natural circulation mode by blocking flow through the CTAH branch. V&V for steady state coupled natural circulation for CIET is reserved for future work, but the CIET educational simulator is still able to give the user a visual experience of what natural circulation is like. This can be seen in Fig. 13 where components within the DRACS loop, Heater branch and DHX branch are visibly hotter than the CTAH branch:

Fig. 13 also shows the ability of the CIET Educational Simulator to be run natively on a Linux gaming laptop. This shows the cross platform capabilities of the CIET Educational Simulator as Figs. 12 and 13 shows its ability to run natively on Windows and Linux systems respectively. A screenshot of the CIET Educational Simulator running natively in MacOS

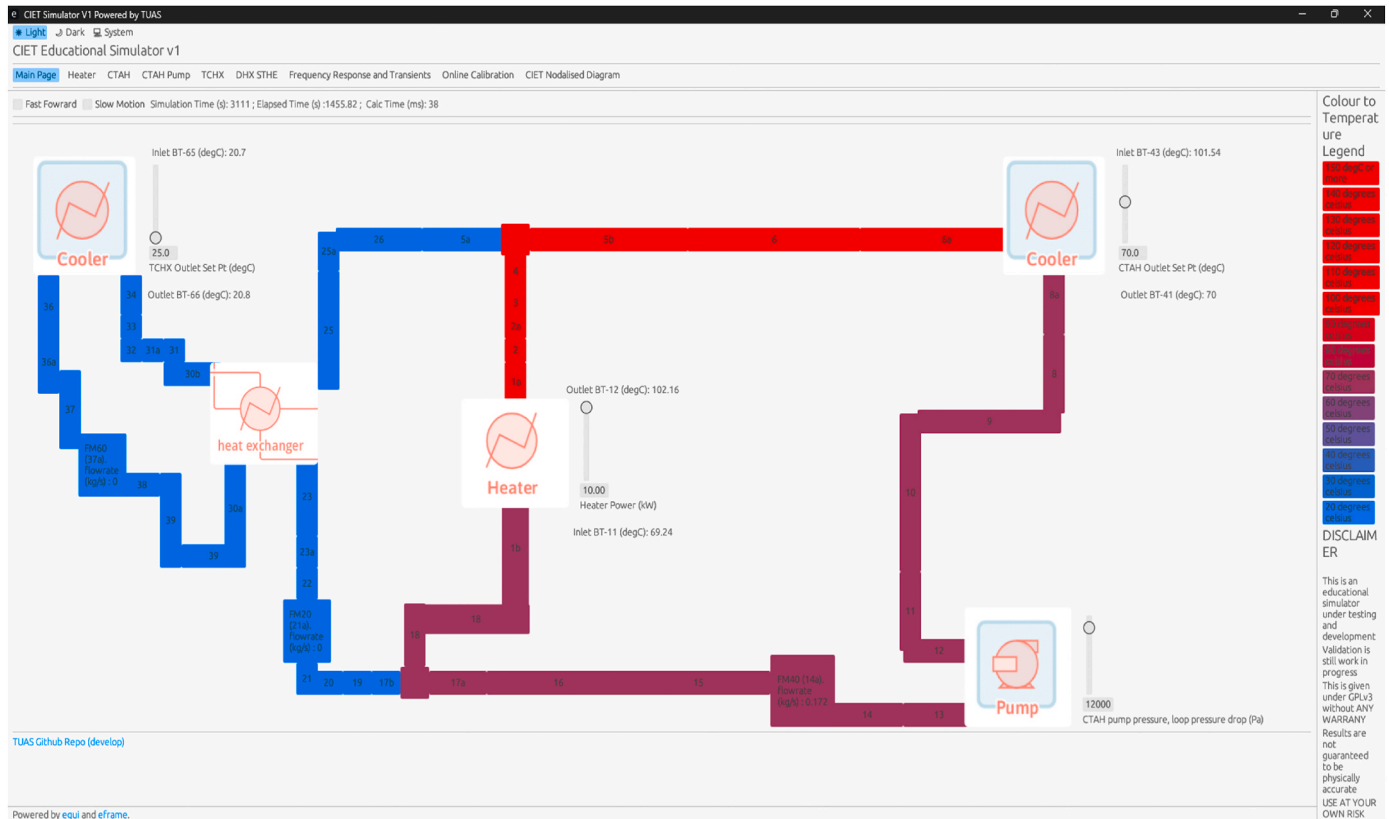


Fig. 12. CIET Educational Simulator Running Natively on Windows in Forced Circulation Mode in Light Mode (Dark Mode also available).

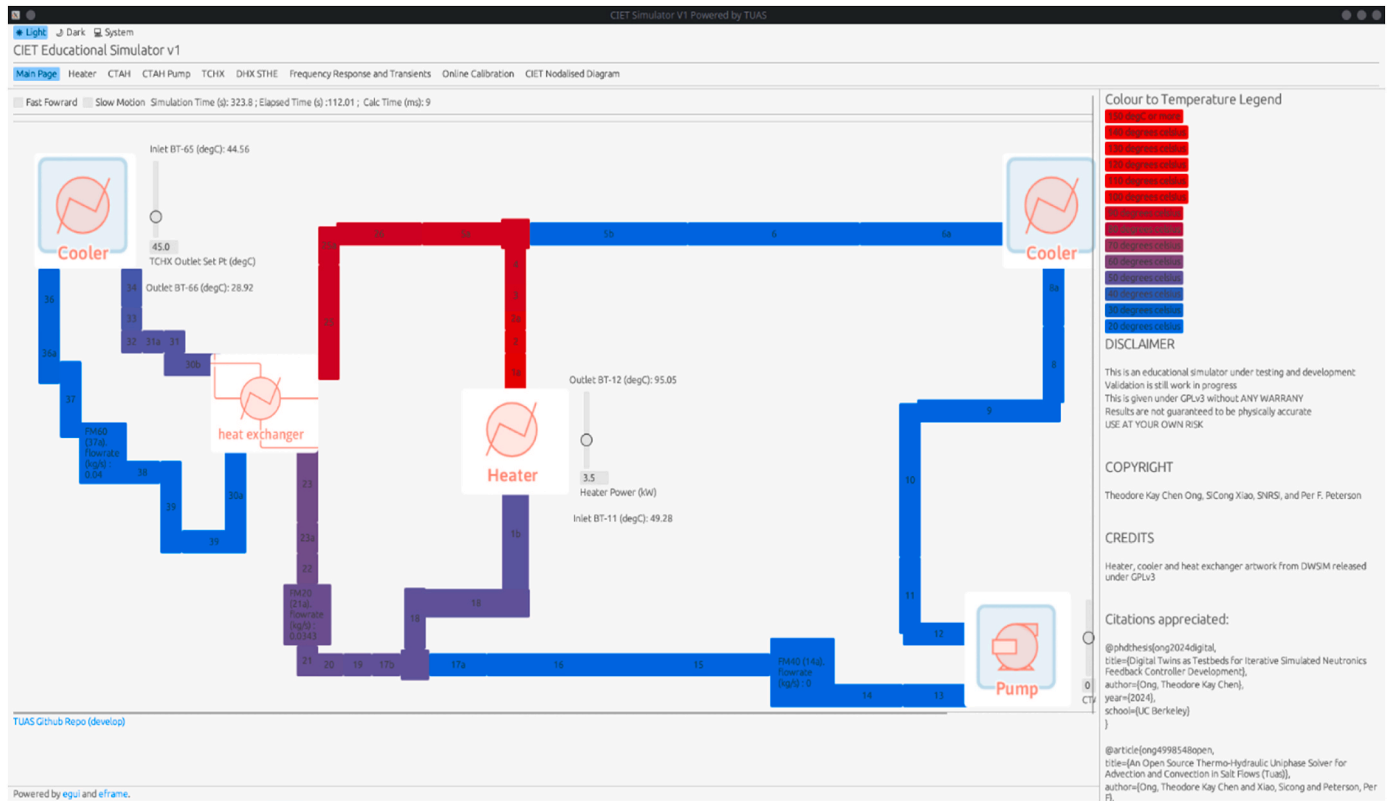


Fig. 13. CIET Educational Simulator Running Natively on an Arch Linux AFTERSHOCK Gaming Laptop in Natural Circulation Mode in Light Mode (Dark Mode also available).

is available in [Appendix A](#). This makes this open-source simulator available to users on all platforms, thus increasing its reach for the purposes of public outreach and education. Moreover, for increased user friendliness and enhanced user experience, the user is also given a “fast-forward” checkbox with which to speed the simulation up. This allows the user to accelerate the simulated system’s transition towards steady state, provided the computing hardware is fast enough.

Besides its use in public outreach and education, the CIET Educational Simulator can facilitate steady state and transient tests without having to write code for different operating conditions for the heater, pumps, TCHX and CTAH. Based on user feedback at the Singapore Nuclear Research and Safety Initiative (SNRSI), it is useful for the user to visualize temperature and heater power evolution with time using graphs. Moreover, it is also useful for the user to plot datapoints for said graphs on a csv file. The CIET Educational Simulator allows for both on the CTAH, CTAH Pump, TCHX, DHX STHE and Frequency Response and Transients Tab. For transient tests such as step response and frequency response as done in SAM validation [18], it is also useful for the user to vary heater power using step and sinusoidal inputs respectively.

4.3.2. Parasitic Heat Loss Solution Verification for InsulatedFluidComponent Objects

CIET contains several insulated pipes for forced and natural circulation. These have some parasitic heat losses associated with them. In the CIET Educational Simulator, InsulatedFluidComponent objects within TUAS were used to simulate these pipes. To ensure that these were programmed correctly, parasitic heat losses through the pipes were compared against an analytical solution, Equation (11). The results are shown in [Fig. 14](#):

From [Fig. 14](#), the parasitic heat loss for the pipes simulated in TUAS agree well with the analytical solution. However, as the pipe grows in length, increases in deviations are observable where the simulated pipes in TUAS have consistently less parasitic heat loss than the analytical solution. This may be due to variation in thermophysical properties, as Equation (11) uses thermophysical properties at a fixed temperature, or due to the axial conduction effect [59]. This axial conduction within the solid piping and insulation is expected to reduce parasitic heat loss because axial conduction was reported to cause performance deterioration in cryogenic counterflow heat exchangers [59].

To deduce the cause of this discrepancy, some investigations were performed. Firstly, thermophysical property variations with temperature and axial conduction were neglected. This resulted in agreement between analytical solution and TUAS to within 0.001 %. From investigations where thermophysical property variations were allowed, the thermophysical property variation only accounted for only a 0.07 K discrepancy between TUAS and the analytical solution. Whereas where axial conduction was turned on, the reduction in parasitic heat loss resulted in about 0.3 K increase of T_{out} . This is consistent with literature which mentions heat exchanger performance reduction due to axial

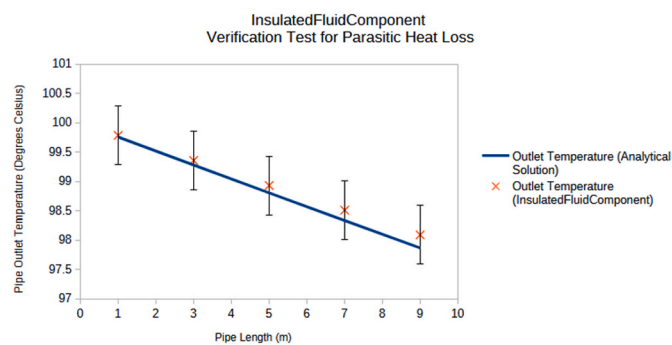


Fig. 14. Parasitic Heat Loss Solution Verification for InsulatedFluidComponent Objects, Error Bars represent 0.5K of Thermocouple Measurement Uncertainty for Reference.

conduction. Additionally, the effect of mesh refinement was also investigated for the 9 m pipe. For a 9 m pipe simulated using 12 nodes, T_{out} was 98.093 °C compared to the 92 node solution where T_{out} was 98.091 °C, a difference on the order of 0.002 K. Hence, mesh refinement is not a significant cause of the discrepancy. Given consistency with observations in literature, and that solution verification for 1D conduction was carried out in previous work [34], one can be reasonably confident that the discrepancies were due to axial conduction and that the TUAS objects for insulated piping were correctly programmed.

4.3.3. Forced circulation step response test

To demonstrate the effectiveness of using the CIET Educational Simulator in simulating transients and to validate the CIET heater model constructed using TUAS, two step response tests were performed using the CIET Educational Simulator. The first step response test was a step up in heater power from 4.24 kW to 6.08 kW and the second was a step down in heater power from 7.15 kW to 5.44 kW.

For both transients, pressure changes across the CTAH pump were set to 13.5 kPa. With this setting, mass flowrates were set at 0.1783 kg/s and 0.1793 kg/s at the start of the transient, which is within 2 % of the flowrate of 0.18 kg/s in the CIET experiment. This discrepancy for steady state mass flowrates given heater power of 4.24 kW–7.15 kW is well within 2 % measurement uncertainty of the Coriolis flowmeters as shown on page 28 of Zweibaum’s dissertation [21]. Unlike SAM, where PID control was used to set flowrates to 0.18 kg/s [36], the pressure drop change across the CTAH pump was set to be constant. Nevertheless, under 13.5 kPa pressure change across the CTAH pump, heater power at steady state of 4.24 kW and 7.15 kW caused less than 2 % variability between the simulated flowrates and the target experimental flowrates of 0.18 kg/s. Therefore, even when step response transients cause the power to change to 6.08 kW and 5.44 kW respectively, the effect of temperature deviations on the mass flowrates are expected to be less than 2 %. Hence, the mass flowrates can be considered approximately constant at about 0.18 kg/s similar to the actual experiment.

Experimental data for both tests were taken from graphs in page 47 of Zweibaum’s dissertation [21]. For each test, two nodalizations of the CIET Heater were used, one with 15 nodes same as the SAM model, and one coarse mesh version using 8 nodes. This investigates the effects of numerical diffusion in TUAS in context forced circulation step response. Results for the step up test, known as test 1, are shown in [Fig. 15](#):

As shown in [Fig. 15](#), the simulated heater outlet temperatures agree with the experimental data for most of the transient within thermocouple measurement uncertainty of 0.5 K shown by error bars. Both simulation and experiment settle to the same steady state at about 100s onwards. This means that the parasitic heat loss calibration was successful. While there is some discrepancy between simulation and experiment, this could be attributed to noise and variability inherent to the experimental data. As shown at about 10s before the start of the transient, the heater outlet temperatures are noisy varying at about 1K. Therefore, the variability observed at about 60s post step response, while beyond the thermocouple measurement uncertainty of 0.5 K, is still reasonable as it is likely within the 1K observed as noise in the

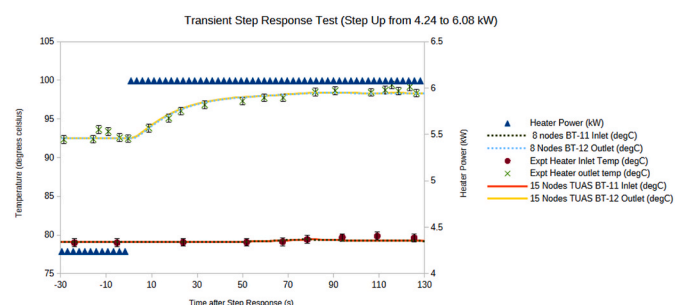


Fig. 15. Step response test 1 for CIET educational simulator heater validation.

experimental data. Of course, one may attribute this extra discrepancy due to nodalisation. However, a mesh refinement from 8 nodes to 15 nodes produces hardly any visible difference in Fig. 15. In either nodalisation, the simulation agrees reasonably well with experimental data. Hence, for this forced circulation transient, a nodalisation scheme of 15 nodes is indeed sufficient. Similar trends are observed in Fig. 16:

In Fig. 16, both nodalisations of the heater follow experimental data of the heater outlet temperature to within the noise of the experimental data. The simulated outlet temperatures for both nodalisations settle to within the same steady state temperature as the experimental data, again within the variations of experimental data. Miniscule deviations between the 8 node heater and 15 node heater can be seen at about 10s post step response. However, these deviations are visibly smaller than the thermocouple measurement uncertainty of 0.5K. This shows that for a mesh refinement from 8 nodes to 15 nodes for the same heater, the effect of numerical diffusion caused by the first order discretization schemes is statistically insignificant since it is smaller than measurement uncertainty. Since this is the case for this forced circulation transient and since the outlet temperature data for both nodalisation schemes follows experimental data closely, using first order discretization schemes with both the 8 and 15 nodes scheme for the heater is reasonably acceptable. However, there are undershoots observed in Fig. 16 as well which are largely absent in the TUAS simulation data. These undershoots are a result of the CTAH PID controller taking time to control the outlet temperature to the set point of 80 °C. This can be seen in the step response for the CTAH inlet and outlet temperatures for both the step up and step down step response transients is shown in Figs. 17 and 18 respectively:

As seen in Figs. 17 and 18, the CTAH outlet temperatures have overshoots and undershoots which deviates significantly from the TUAS simulation. Zou experienced a similar phenomenon, mentioning that this may be due to the difference in PID controller tuning between the SAM simulation and CIET experiment [18]. The validation study for SAM left the implementation of realistic control of the CTAH and TCHX in future work. For this work, this approach was also taken as PID controller tuning for the CTAH was not meant to match experimental data, but to ensure a constant outlet temperature within a reasonably short timeframe for a user-friendly experience. These undershoots and overshoots in heater inlet temperature do not occur until after about 80s of simulation time as seen in Fig. 16 due to the time it takes for a thermal pulse to traverse the forced circulation loop. Hence, at least up till 80s, the heater transients for both inlet and outlet temperatures match well with experimental data.

For the CTAH in both step responses, the simulated temperatures are insensitive to whether the 8 node or 15 node first-order nodalisation was used for the heater. This is unlike the case for natural circulation transients, where the driving force for flowrates were strongly coupled to temperature profiles, and therefore nodalisation schemes. Additionally, the CTAH inlet temperatures generally agree well with experimental data, except that in the step-up step response in the region of 30s–80s post step response, the TUAS simulated CTAH inlet temperatures are

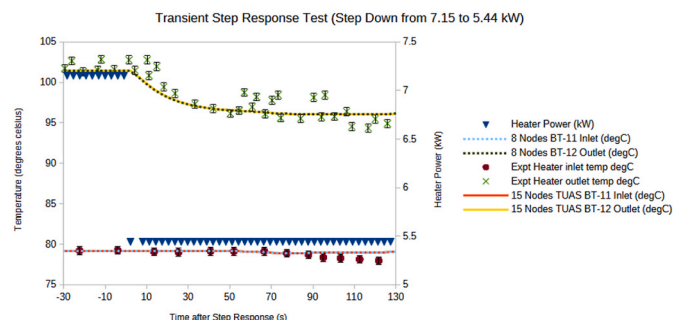


Fig. 16. Step response test 2 for CIET educational simulator heater validation.

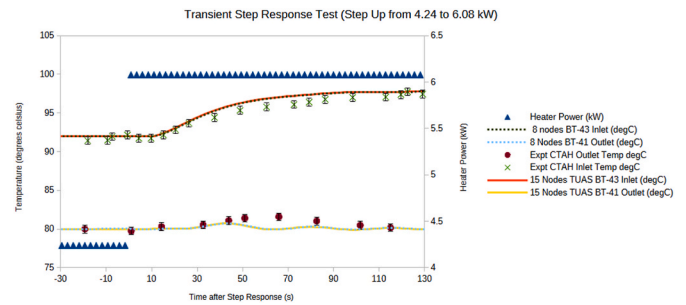


Fig. 17. Step response test 1 CTAH validation.

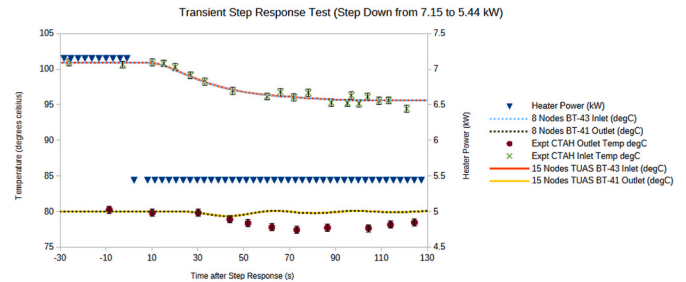


Fig. 18. Step response test 2 CTAH validation.

slightly higher than the experimental data (about 1.5K). For this first published iteration of the CIET Educational Simulator, no calibration has yet been done for the forced circulation loop. The only difference between this and the SAM model is the addition of two small mixing nodes as described earlier. Yet even in this initial model, the CTAH inlet temperatures simulated by TUAS seemed to match the experimental data reasonably well. Given this success, the 15-node scheme was eventually used within the CIET Educational Simulator to be consistent with the SAM nodalisation. However, users are given the choice to select the 8-node heater within the “Online Calibration” tab in Fig. 12 if they wish to repeat the simulated step response tests in this work. Future work can include calibration the thermal inertia of the loop to better match experimental data, and then a fuller V&V study for forced circulation transients can be performed.

The success of the step response validation tests demonstrated that the CIET Educational Simulator can simulate forced circulation transients for CIET in real-time. As evidenced by Figs. 12 and 13, a timestep of 0.1s took 38 ms and 9 ms respectively to calculate. Hence, it has demonstrated the potential to simulate other forced circulation transients in future since mass and energy balances for all components in CIET were solved in less than 0.1s. The calculation times may vary, but on an Arch Linux gaming laptop with a i7-10875H CPU purchased in June 2020, the calculation times are on average faster than the simulation timesteps. This even allowed for faster than real-time simulation. With more modern CPUs on desktops currently available, faster calculation times are expected. Hence, there is a potential for the CIET Educational Simulator to run in real-time with even shorter timesteps. This suggests that real-time forced and natural circulation simulation transients could be performed using TUAS in future with suitable hardware, provided that suitable validation datasets for such transients are available.

5. Conclusions

In this work, the open source TUAS library solver was constructed and its underlying concepts were explored. The TUAS library was also demonstrated in its capability to construct GUI simulators like the CIET Educational Simulator and which could simulate forced circulation

transients for CIET in real-time, and could, with suitable validation studies in future, demonstrate the possibility of simulating transients for CIET in real-time and natural circulation flows as well. TUAS has also demonstrated its capability to accurately perform steady state simulations of simulate molten salts and natural circulation loops with the possibility of real-time simulation in future given suitable validation datasets. This open-source structure as well as real-time simulation capabilities of the TUAS library makes a significant contribution to an otherwise sparse open-source GUI simulation ecosystem for FHRs in 2024–2025. To ensure that TUAS could be relied upon, models of the isolated DRACS natural circulation loop within CIET and the shell and tube heat exchanger were constructed using the TUAS library for verification and validation efforts. The isolated DRACS loop agreed well with simulated data from SAM and the shell and tube heat exchanger agreed well with experimental data. This demonstrated the ability of TUAS library to simulate steady state natural circulation loops with oil and forced convection heat transfer with molten salt even with first order schemes. Moreover, the CIET Educational Simulator powered by TUAS, showed its capability in providing a user friendly experience for validation tests as shown in the step response validation of the CIET Heater with experimental data. Hence, the initial V&V efforts were successful.

This work provides an open-source thermal hydraulics library for the FHR and other molten salt reactors. For this work to continue, the TUAS library should be validated against experimental natural circulation data, both in CIET and in molten salt natural circulation loops. This includes further calibration and V&V studies of the CIET Educational Simulator. If this is successful, TUAS library could be validated using transient natural convection data. It is even possible that TUAS library will eventually be used to simulate molten salt reactors such as the molten salt reactor experiment (MSRE) given a point kinetics model, suitable neutron precursor drift model and/or some artificial intelligence based surrogate neutronics model. Moreover, the GUI of the CIET Educational Simulator could be improved in terms of readability and aesthetics to improve its user-experience especially for public education.

To further improve the accuracy of TUAS, implementation of higher order discretization schemes can be explored in future. Of the schemes that can be explored, higher order schemes in literature that avoid the oscillatory behaviour of second order schemes and improve upon the numerical diffusion issues in first order schemes [60] are of interest. These can be introduced in future iterations of TUAS where users are allowed to select their preferred discretization scheme as is done in OpenFOAM [61]. Moreover, if a more rigorous V&V effort is desired, systematic approaches in literature, including Bayesian methods [62], can also be explored in future work.

A more ambitious outlook would include coupling of the TUAS library to new single phase compressible and multiphase flow solvers

written in Rust so that boiling crisis can be simulated in the context of FHR startup. Moreover, 2D and 3D GUI simulators of the FHR could be constructed with different kinds of secondary loops in future for the purposes of outreach, education and research. However, this would include significant amounts of work as boiling would require complex equations and consideration of liquid properties such as surface tension [63]. Despite this challenge, it would be an interesting endeavour because natural circulation instabilities are complex phenomena to study and simulate [64]. Finally, TUAS could potentially be expanded to model flows using a 2D or 3D modelling so as to model local flow recirculation which could be important in full core modelling or where large tanks of salt are stored.

All these efforts could cumulate in a richer open-source ecosystem for conducive learning and research in nuclear reactor safety simulation.

CRedit authorship contribution statement

Theodore Kay Chen Ong: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Sicong Xiao:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Per F. Peterson:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Per F. Peterson reports a relationship with Kairos Power LLC that includes: employment and equity or stocks. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

English Alphabet A to J

A_{XS}	Cross Sectional Area m^2
c_p	Specific Heat Capacity $\frac{J}{kg \cdot K}$
d_e	Hydraulic Diameter (m) Specifically for Tube side of Shell and Tube Heat Exchanger
d_h	Hydraulic Diameter (m)
d_i	Inner diameter (m)
d_o	Outer Diameter (m)
D_e	Hydraulic Diameter specifically for Shell Side of Shell and Tube Heat Exchanger (m)
D_i	Inner Diameter specifically for Shell Side of Shell and Tube Heat Exchanger (m)
D_o	Outer Diameter specifically for Shell Side of Shell and Tube Heat Exchanger (m)
$f_{darcy} \cdot f$	Darcy Friction Factor

(continued on next page)

(continued)

g	Earth's Gravitational Acceleration $\frac{m}{s^2}$
h	Convective Heat Transfer Coefficient $\frac{W}{m^2K}$
$h_{enthalpy}$	Specific Enthalpy (J/kg)
h_{shell}	Convective Heat Transfer Coefficient $\frac{W}{m^2K}$ at shell side of Shell and tube Heat Exchanger
h_{tube}	Convective Heat Transfer Coefficient $\frac{W}{m^2K}$ tube side of Shell and Tube Heat Exchanger
H	Thermal Conductance (W/K)
$H_{lateral,i}$	Thermal Conductance for lateral surface i $\left(\frac{W}{k}\right)$

English Alphabet K to Z

K	Form Losses
l	Length of component (m)
$\ln$	Natural Logarithm
$\dot{m}$	Mass Flow Rate (kg/s)
Nu_{shell}, Nu_s	Nusselt number on the shell side of the Shell and Tube Heat Exchanger (STHE)
P	Power (W)
ΔP	Pressure Difference (Pa)
$\Delta P_{change\ branch}$	Pressure Difference (Pa) across a branch of Serially connected Fluid Components
$\Delta P_{source,i}$	Pressure Source (Pa) for Fluid Component i
Pr_i	Prandtl Number evaluated at Bulk Fluid Temperature
$Pr_{i,shell}$	Prandtl Number evaluated at Bulk Fluid Temperature for Shell Side of Shell and Tube Heat Exchanger (m)
Pr_w, Pr_{wall}	Prandtl Number evaluated at Wall Temperature
$Pr_{w,shell}$	Prandtl Number evaluated at Wall Temperature for Shell Side of Shell and Tube Heat Exchanger (m)
q	Heat Flux $\frac{W}{m^2}$
$q_j^t A_j$	Heat Flux at time t multiplied by area for face j
$\dot{Q}$	Heat Flow Rate (W)
$Re_s, Re_{f,shell}$	Reynold's number on the shell side of the Shell and Tube Heat Exchanger (STHE), evaluated at bulk fluid temperature
t	Time (s)
Δt	Timestep (s)
T_x^t	Temperature at position x and time t (K)
u	Fluid Superficial Velocity $\frac{\dot{m}}{\rho A_{xs}}$ (m/s)
x	Length in x direction (m)
Δx	Distance between centers of 1D control volumes
z_i	Height Change (m) for component i

Greek Symbols

ρ	Density $\frac{kg}{m^3}$
λ	Thermal Conductivity $\frac{W}{m\ K}$

Controller Nomenclature

α	Derivative Filter Ratio
$\bar{e}(s)$	Dimensionless error term
$G(s)$	Transfer Function (dimensionless)
h_{TCHX}	Simulated Convective Heat Transfer Coefficient $\frac{W}{m^2K}$ at surface of the TCHX
K_c	Controller Gain (dimensionless)
$\bar{O}(s)$	Dimensionless output term
s	Complex Frequency (Hz)
τ_d	Derivative time (s)
τ_I	Integral time (reset time) (s)

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jandt.2025.03.006>.

Appendix A. Cross Platform Capability

The CIET Educational Simulator is able to run on Windows, Linux and MacOS. Figs. 12 and 13 show the CIET Educational Simulator running natively in Windows and Linux respectively. Whereas Figure A1 shows the simulator running on MacOS.

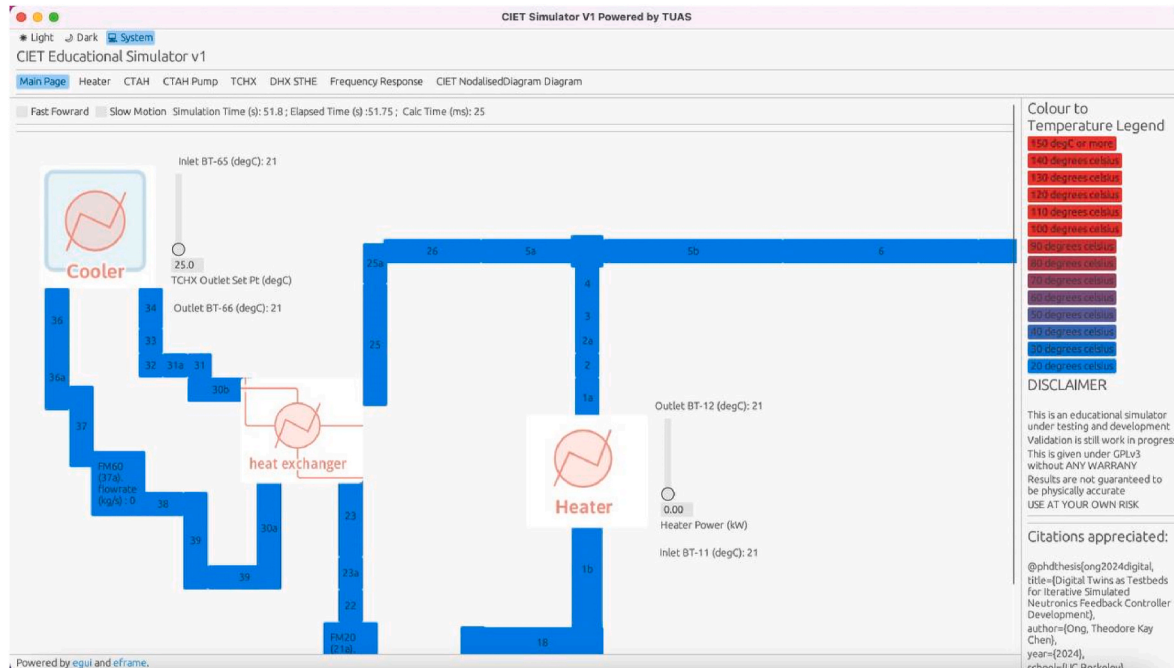


Fig. A1. Early Iteration of CIET Educational Simulator on Running Natively on MacOS (Light Mode)

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